

Manual adjustment for threshold value

1. Press the  button.

▶ The threshold value is displayed, and the display blinks.



2. Press the  button.

▶ The threshold value is changed.

3. When proper threshold value is displayed, press the [Teaching/Enter button].

▶ It returns to the RUN mode.

NOTE

- The standard threshold value is set to 50.

4.9.4.2 Vacuum Pressure Switch Sensor

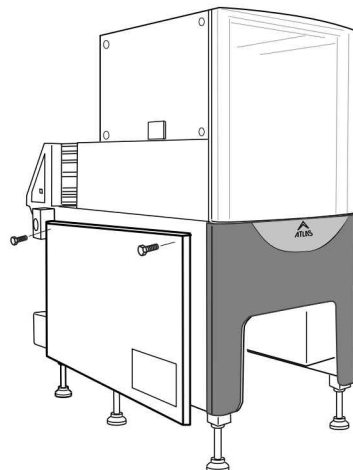
NOTE

- The * symbol used with the serial number shown on the board represents the board version number.

1. Turn the main power OFF.
2. Remove left lower cover.
3. Remove the XJ9 connector from the SCU2 (P-5492 *) board.

NOTE

- The connector of the vacuum pressure switch sensor cables have the following numbers:
Connector Number (XJ9) : Color
Pin 2 : Brown
Pin 3 : Blue

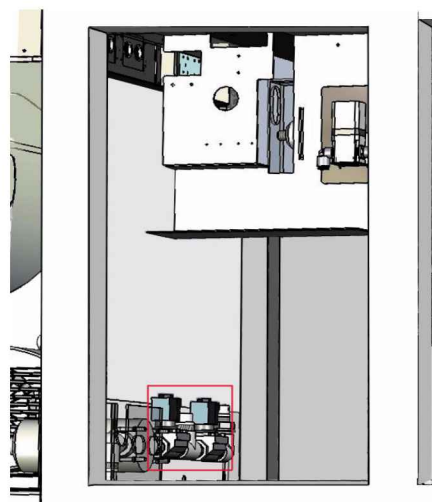


4. Remove the vacuum pressure switch sensor.



CAUTION

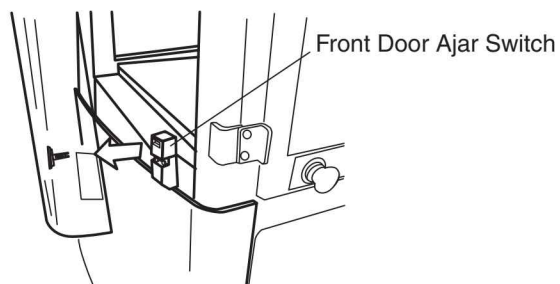
- **Make sure that the cable is inserted properly. If the machine is operated when the connect is substandard or the terminal is incorrect, the sensor can malfunction and / or damage can be caused.**



5. Install the vacuum pressure switch sensor.
6. Reinstall all parts in the reverse order of removal.

4.9.4.3 Front Door Ajar Switch

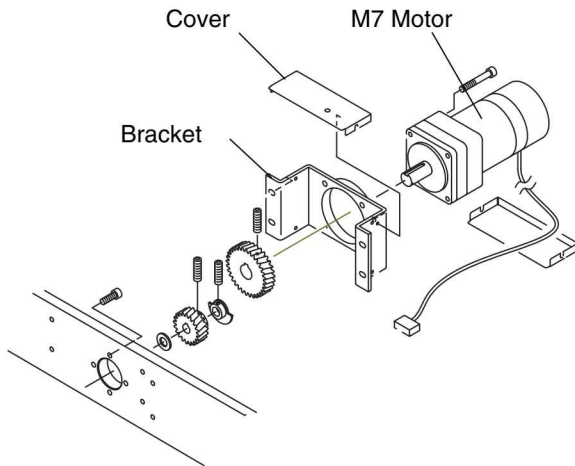
1. Turn the main power OFF.
2. Open the front cover.
3. Remove the front door ajar switch and bracket.
4. Loosen the screw at the bottom of the switch, and remove the cable.
5. Reattach the cable to the new switch.
6. Install the new switch into the main unit.
7. Reinstall all parts in the reverse order of removal.



4.10 Shaker Unit (Option)

4.10.1 Replacing the M7 Motor

1. Turn the main power OFF.
2. Open the front cover.
3. Disconnect the two cables that are connected from the M7 servo driver to the M7 motor.
4. Remove the cover.
5. Loosen the allen screws, and remove the M7 motor from bracket.
 - M6 set bolt
 - Tool 5 mm hex wrench



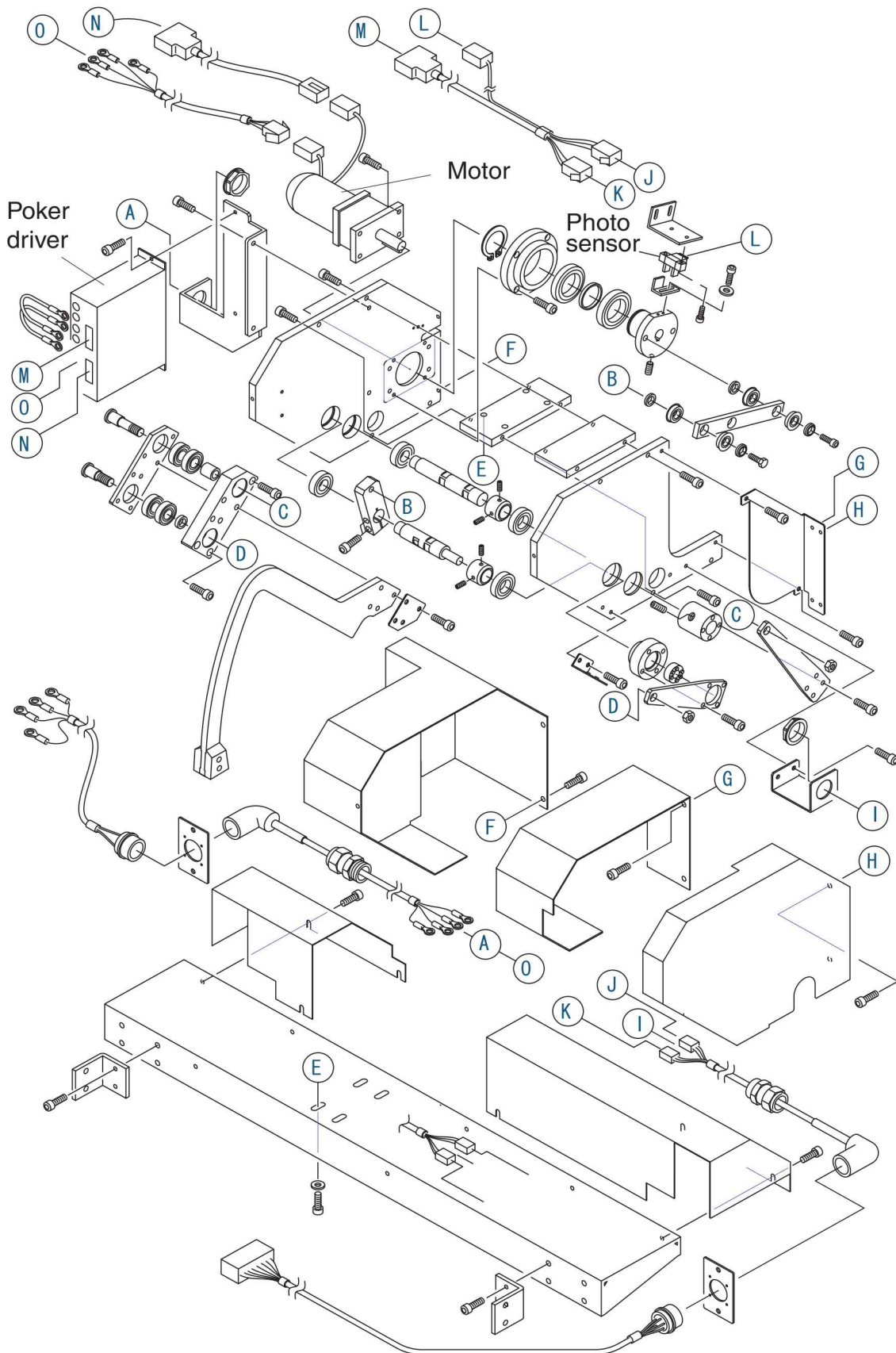
6. Attach the original gear to the new motor; then apply a small amount of graphite grease.

NOTE

- Replace the motor only. The gears are not included.

7. Install the new motor.
8. Reinstall all parts in the reverse order of removal.

4.11 Poker Unit (Option)



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5 APPENDIX

5.1 Program Installation Procedure from Web-RCU

CAUTION

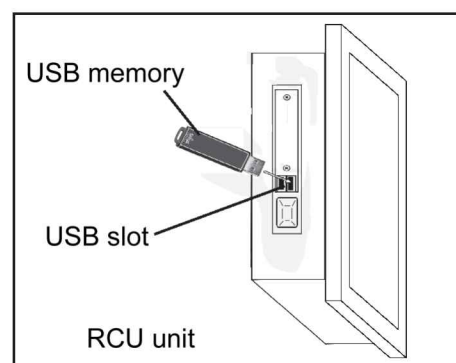
During the software installation or backup process, never turn off the power or remove the installation card.

NOTE

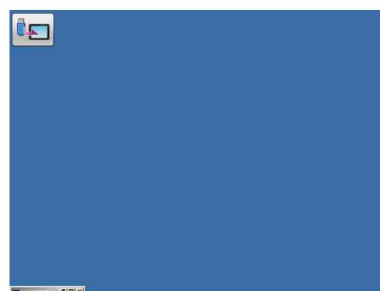
During installation or backup, the data is overwritten between the RCU and the main board. If the process is interrupted, the installation card may be broken, and the machine communication function may be disabled to restore. Wait while the screen in-process is displayed. Turn off the power or remove the installation card only after the process is completed.

5.1.1 Installation Software Start up

1. To the USB slot of RCU unit, insert a USB memory to which the installation software is written. Turn ON the main power.
2. An initial screen and the startup screen for Windows XP appear.

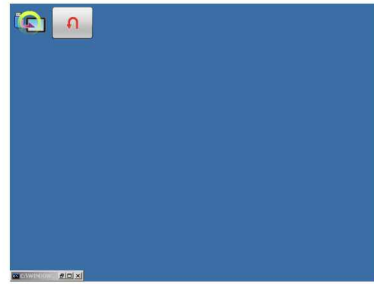


After that, the  key is displayed for five seconds. When this key is not touched, normal application starts up.



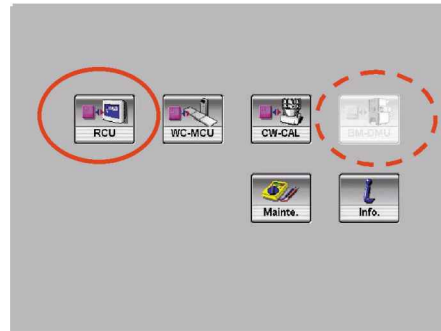
3. Touching the  key. A screen for “Install Drive Check” appears (Displayed for approximately 20 seconds at maximum).

To cancel installation, touch the  key. The normal application starts up.



- When the installation software starts up, the main menu screen shown as below appears. By touching these buttons on the main menu, various menus appear to enable the operations.

In the example in the figure below, when the  key circled in a solid line is pressed, the main menu is switched to the menu to install or backup the RCU software. When the software to write in the installation card is not inserted, the key on the menu is faded as the area circled in a dashed line.



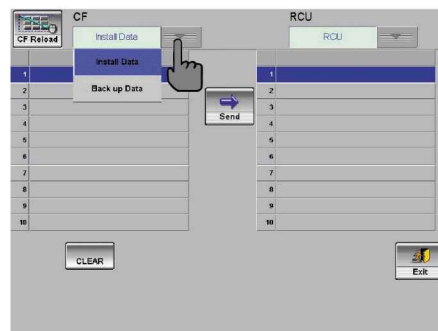
CAUTION

Before starting the software installation, be sure to backup the existing software to the installation card. Once the new software is written, the previous status cannot be restored without the installation card storing the previous software.

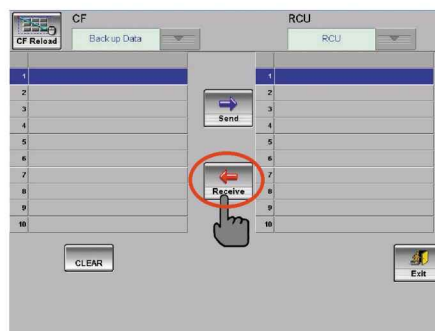
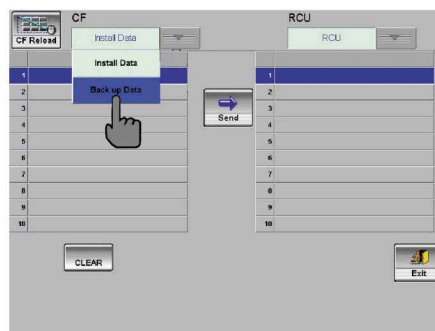
5.1.2 RCU Software Backup

Before installing the new software, backup the written RCU software to the CF card.

- Press the  key on the main menu screen.



- The installation screen appears. Press the CF drop-down key at the upper left of the screen. Select "Back up Data" from the list, and press the "Back up Data" key.

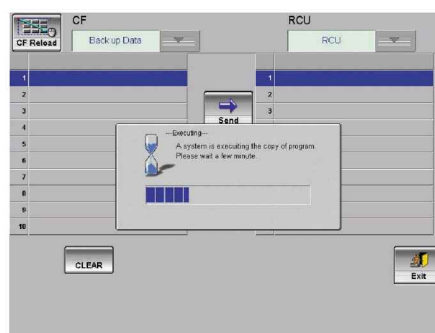


- The  key as shown in the above figure appears. When the  key is pressed, the confirmation dialog screen appears to confirm whether or not to backup the RCU software.



- Press the  key to backup.
Press the  key to return to the previous screen.

- Press the  key on the confirmation screen to start the backup process. The backup process screen appears.
The waiting time is approximately 5 minutes, depending on the software.



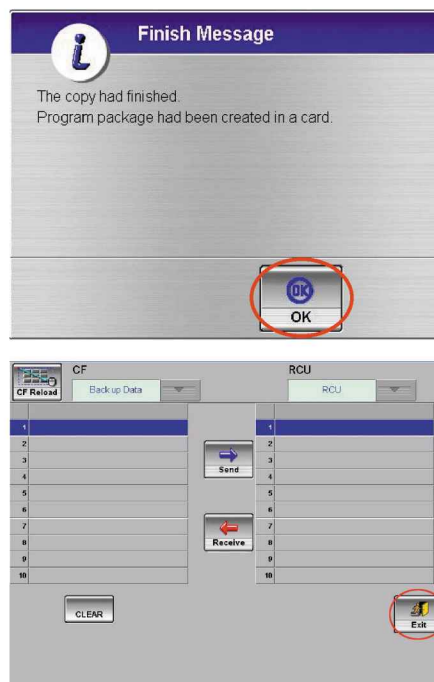
CAUTION

During the backup process, never turn off the power or remove the installation card.

6. When the backup process is completed, the Finish Message appears.

7. The RCU backup is completed. Press the  key to return to the previous installation screen.

8. Press the  key on the installation screen to return to the main menu screen.



5.1.3 RCU Software Installation

When the stored RCU software does not need the backup or when the backup process is completed, install the RCU software. The following two methods can be applied for the RCU software installation.

1. Install Software to Existing Model

Install only the new software while keeping the existing parameter files for RCU.

2. Install Software Newly

Install the new software and the parameter file from the card when the parameter file for RCU is not existing or deleted.

Unless otherwise indicated, install the RCU software by the method "5.1.3.1. Installing Software to Existing Model" after the factory shipment.

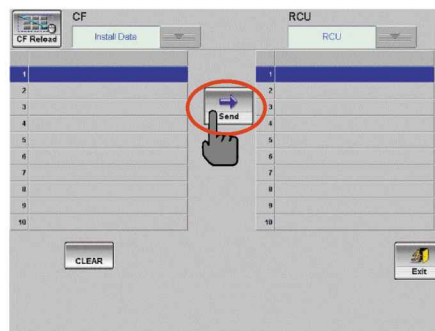
Follow "5.1.3.1. Installing Software to Existing Model" after the existing parameter file for RCU is deleted.

The RCU parameter file includes the settings (displayed language, back light control time, etc.) made on the RCU control menu screen and the shot product images.

5.1.3.1 Installing Software to Existing Model

Install only the new software while keeping the existing parameter files for RCU.

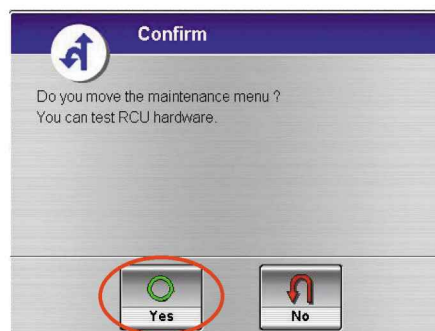
1. Press the  key on the main menu screen.
2. The RCU installation screen appears. Press the  key.



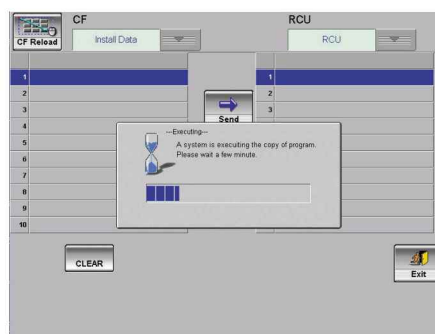
3. The confirmation dialog screen appears to confirm whether or not to install the RCU software.

4. Press the  key to install.

Press the  key to return to the previous screen.



5. When the  key is pressed on the confirmation screen, the installation process starts, and the installation process screen is displayed.
The waiting time is approximately 5 minutes, depending on the software.



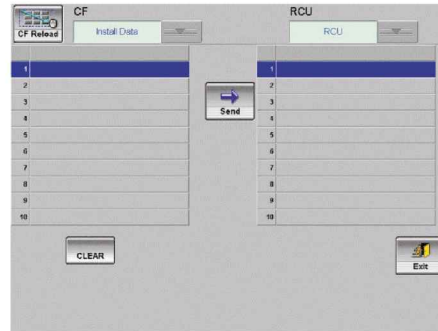
CAUTION

During the installation process, never turn off the power or remove the installation card.

6. When the backup process is completed, the Finish Message appears.
7. The RCU installation is completed. Press the  key to return to the previous installation screen.



8. Press the  key on the installation screen to return to the main menu screen.
9. To reflect the result of software installation, turn off the power, remove the installation card from the slot, and turn ON the power again.



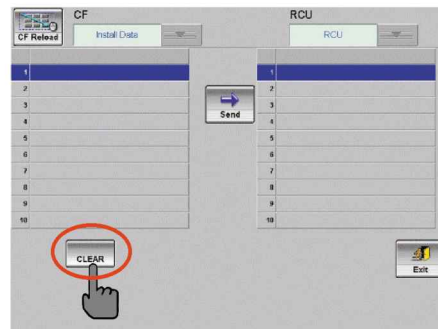
5.1.3.2 Installing Software Newly

Install the new software and the default parameter file from the installation card after the existing parameter file for RCU is deleted.

TIP

- Install the RCU software by the method "5.1.3.1. Installing Software to Existing Model" after the factory shipment unless otherwise indicated.
- Select "5.1.3.1. Installing Software to Existing Model" after deleting the existing parameter file for RCU.

1. Press the  key on the main menu screen.
2. The RCU installation screen appears. Press the  key.



3. The confirmation dialog screen appears to confirm whether or not to delete the RCU parameter file.

Press the  key to delete the parameter file.

4. After deleting the parameter file, install the new software by referring to "5.1.3.1 Installing Software to Existing Model". When the existing parameter file is deleted, the default parameter file is automatically copied from the installation card.



ATLAS-204 Replacement procedure for RCU, BM-DMU, and the Main Software

A1. Overview

This manual describes how to replace the software for RCU, BM-DMU, and the main unit (MCU1, MCU2, SCU1, and SCU2).

When replacing the main software, the write procedure for the main software "Industrial Machinery 05-00059_write procedure for main software.doc" is necessary, in addition to this procedure manual.

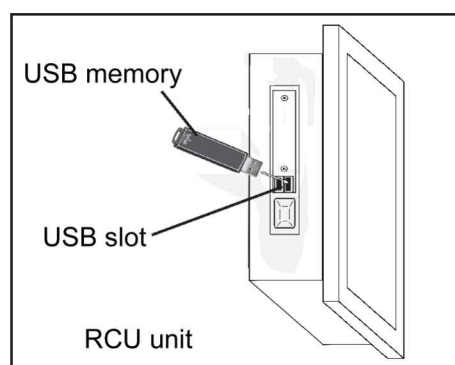
A2. USB Memory to be used and Storing File

When replacing RCU or DMU Software, the Installer Program File (W0002F) must be stored in the USB Memory, along with the file of each program.

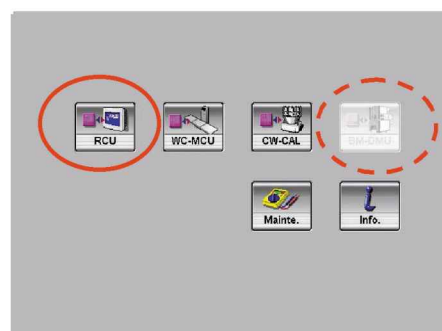
The folder for storing the program of RCU	RCU
The folder for storing the program of BM-DMU	BMDMU
The folder for storing the main unit program	BMFILE

A3. Replacement Procedure for RCU and BM-DMU Software

1. Turn off the main power before starting operation.
2. Insert the USB Memory into the slot of the ATLAS remote control.



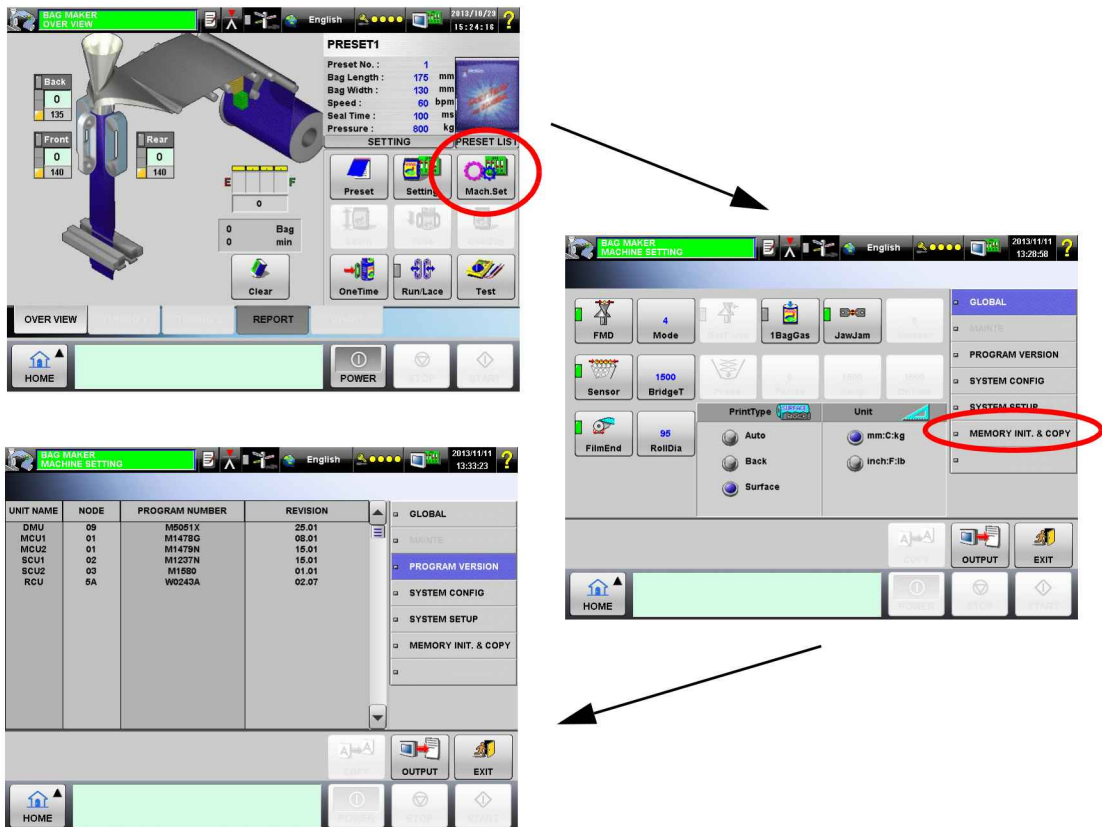
3. Turning on the main power starts the installation screen.



4. Press the [RCU] key as shown above to install the software for RCU.
5. Press the [BM-DMU] key as shown above to install the software for BM-DMU.
6. Remove the compact flash card from the slot and turn on the main power again.

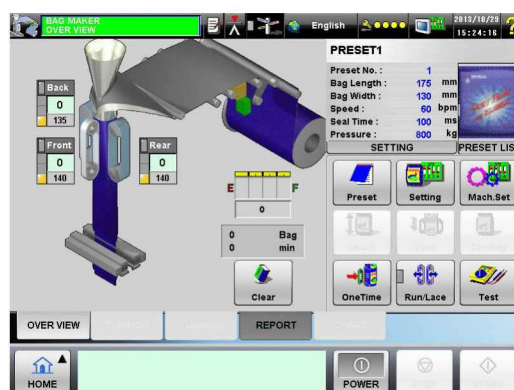
When the remote screen is displayed, press the [Mach Set] Key on the OverView screen.

Then press the [PROGRAM VERSION] key to confirm the program number.

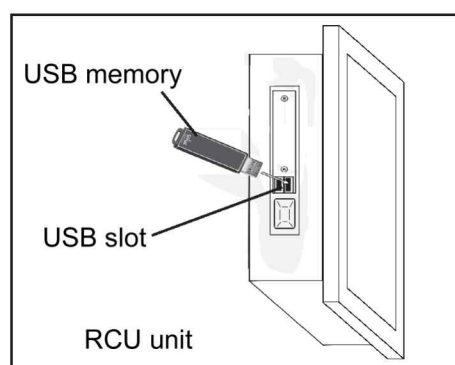


A4. Replacement Procedure for the Main Software

1. Turning on the main power before starting operation to display the operation screen of the bag maker.



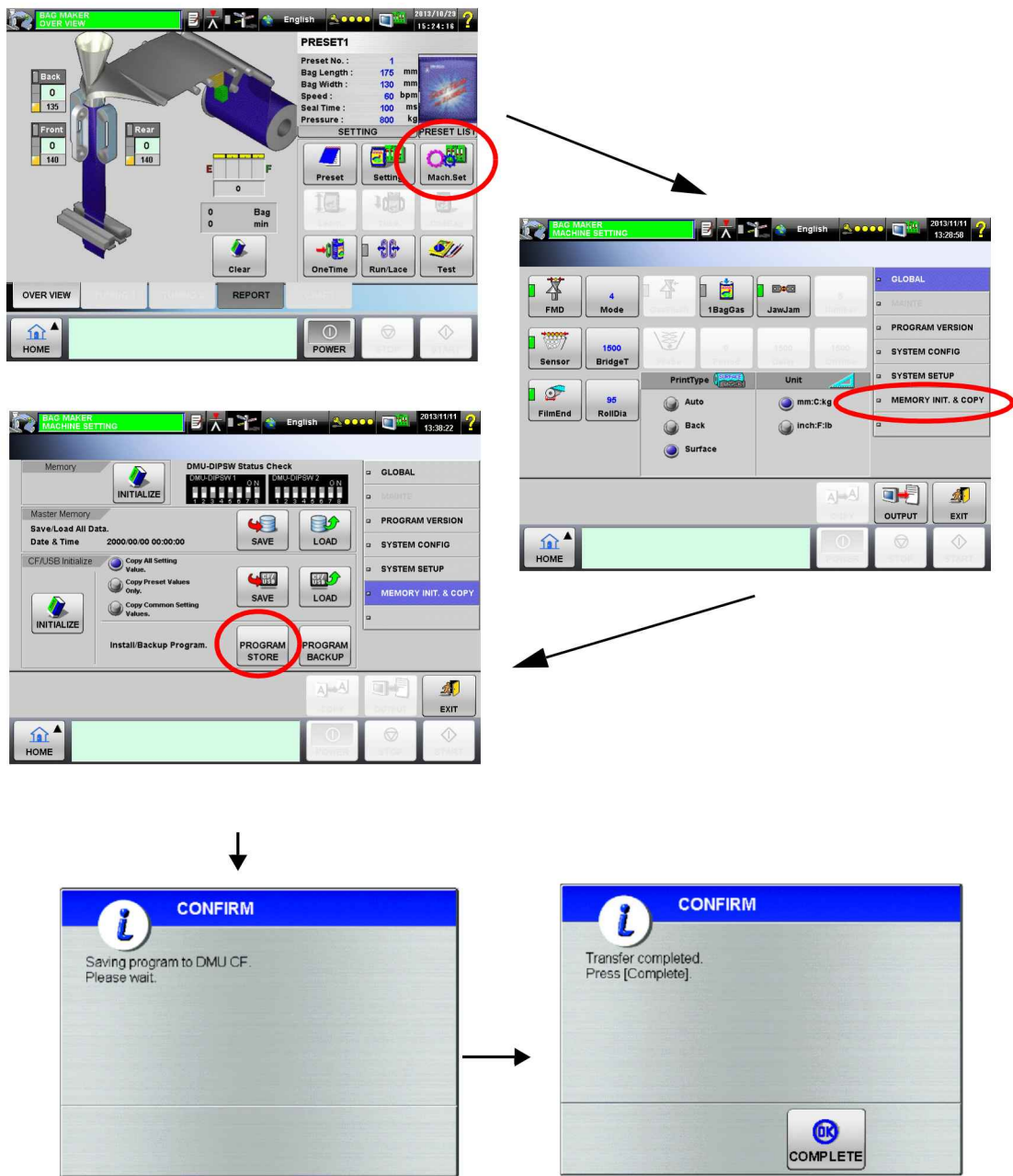
2. Insert the USB Memory into the slot of the ATLAS remote control.



3. Set the operation level to the Maintenance level.

Press the [Mach Set] key on the OverView screen and press the [MEMORY INIT & COPY] key on the next screen.

Then press the [PROGRAM STORE] key on the Memory Init & Copy screen.



When reansfer is completed, refer to the write procedure for the main software "Industrial Machinery 05-00059 write procedure for main software.doc" for main software writing.

5.1.4 Errors during Installation and Backup

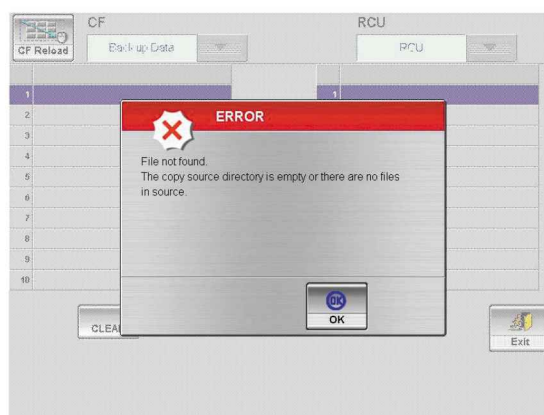
5.1.4.1 Error Display Screen

During the installation or the backup process for the RCU software, the error messages as shown below may appear.

- When an error occurs during the installation or the backup process for the RCU software, press the



key to return to the previous screen and handle the error by referring to the below table.



Error during installation or backup of RCU software

Table 5-1 Countermeasures for Error during Installation or Backup

Trouble	Cause	Countermeasure
Necessary screen buttons are not displayed at the installation or backup.	1.The "program.inf" and program name are not matched. 2.The sum value of "program.inf" is incorrect. 3.The program for the installation is not included in the installation card.	Use the installation card including the correct programs.
The operation is paused after the following messages appear in the error dialog screen. - File not found. - The copy source directory is empty or there are no files in source.	1.There is no data to be correctly installed in the installation card. 2.The backup process is performed for the RCU program while the program does not exist in the RCU.	Use the installation card including the correct programs. The backup process cannot be performed without the existing program.
The operation is paused after the following messages appear in the error dialog screen. - Network connection error. - Can not connect to main board. - Reboot this Machine.	The Ethernet is not connected or improperly connected between the RCU and the main unit.	Pause the installation operation, and check if the machine starts up correctly. Without establishing the network, installation or backup of the main program cannot be performed.

Table 5-1 Countermeasures for Error during Installation or Backup (Continued)

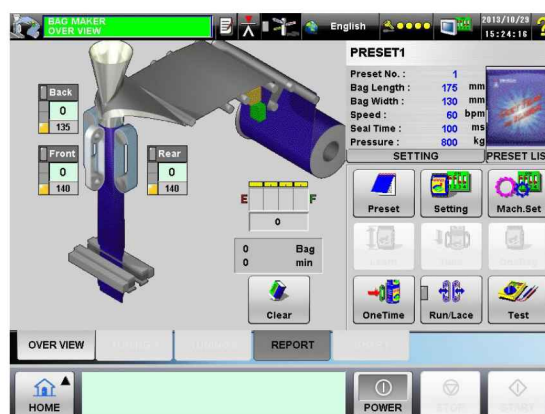
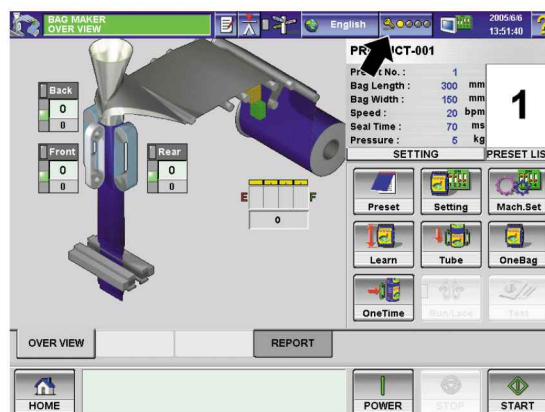
Trouble	Cause	Countermeasure
The operation is paused after the following messages appear in the error dialog screen. - Data send error! - Reboot this Machine.	"40 PCCard access error." Since the CF card does not exist on the main board, the error occurs during the installation and backup.	Insert the CF with the appropriate program to the main board.
	**** download name check error." or **** download version check error." or **** download sum check error." The program name or sum value defined in the *** program does not match "program.inf". The program with an incorrect version may be tried to be installed.	Check the installation data.
The operation is paused after the following messages appear in the error dialog screen. - Data send error! - Reboot this Machine.	**** start address error." or ****program download error." The contents in the *** program cannot be read. The installation data may be broken, or a different program type is operated.	Check the installation data.
	"73 *** DIPSW error." The DIP switch setting is incorrect.	Check the DIP switch on the *** board.
The error messages other than those above appear in the message box.	Errors occur during the internal process.	After rechecking the installation data and no error is found, inform the recorded error message to the manufacture.

5.1.5 Maintenance level

When switching the operation level to the Maintenance level, the password for the Maintenance level is required.

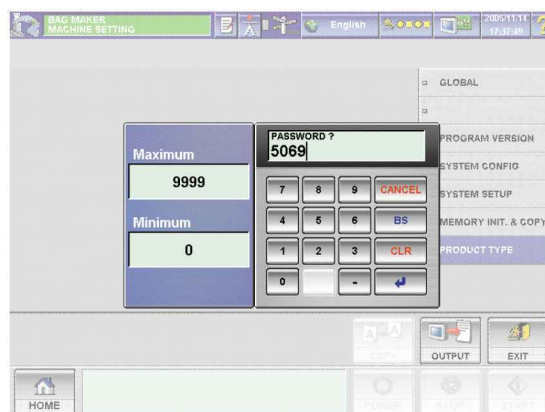
Follow the procedure to switch to the Maintenance level.

1. Press the operation level key  on a screen.
 - ▶ The screen to select the operation level will appear.
2. Select the Maintenance level.
 - ▶ The keyboard to enter the password will appear.
3. Enter the password for the Maintenance level.
 - ▶ The Operation Standby screen for the Maintenance level will appear.
4. Press the  key.
 - ▶ The MACHINE SETTING screen will appear.



5.1.5.1 SYSTEM CONFIG

1. Press the [SYSTEM CONFIG] index.
 - ▶ The keyboard to enter the password will appear.



- Enter the password "5055+(date×2)"= four figures.

►The SYSTEM CONFIG screen will appear.



5.1.5.2 Memory Initialization

NOTE

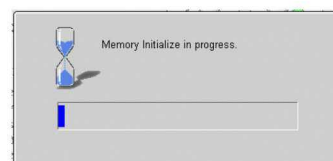
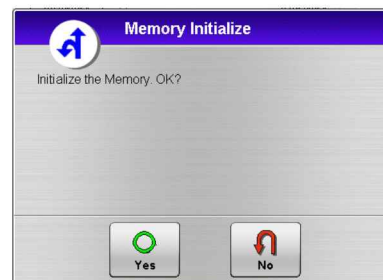
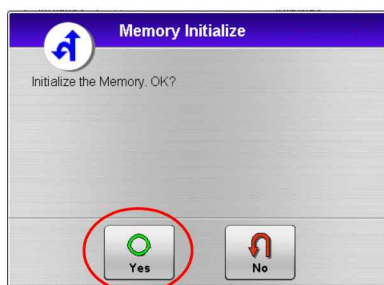
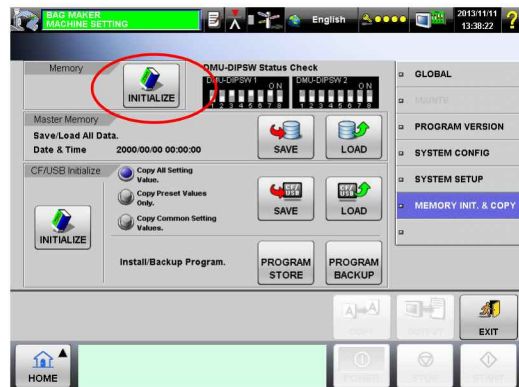
- Before executing memory initialization, ensure to perform [Save All Setting Values] to the USB memory for backup purpose.

- Press the  key of [Memory].

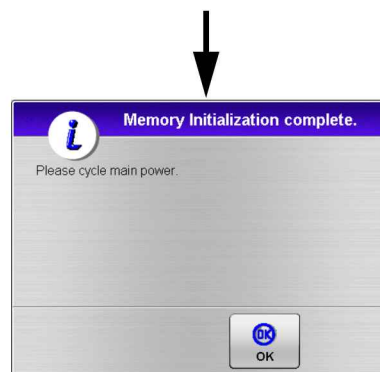
The confirmation screen appears.

- Press the  key to initialize the memory.

To cancel restoration, touch the  key, and the screen returns to the previous screen.



- When memory initialization completes, a message "Turn On the power again." appears.
Touch the  key, turn off the power according to the guidance, and restart the system. (Memory initialization is executed at restart.)



5.1.5.3 Master Memory

Master Memory contains the setting data according to the machine specification (all setting data), and is saved in the CF memory at factory shipping.

When the setting values as "Machine set values" or "Preset values" need to be returned to the original data after changed, Master Memory is required to be restored.

In saving the setting data in the CF memory after setting new "Machine set values" or "Preset values", save these values in Master Memory.

Restoration from Master Memory

TIP

- Even when wrong setting has been made by user or memory initialization has been mistakenly performed, [Restore All Setting] in [Master Memory] enables to return the data to the setting data when [Save All Setting to Master Memory] is last executed.

- To restore data from Master Memory, touch the



key.

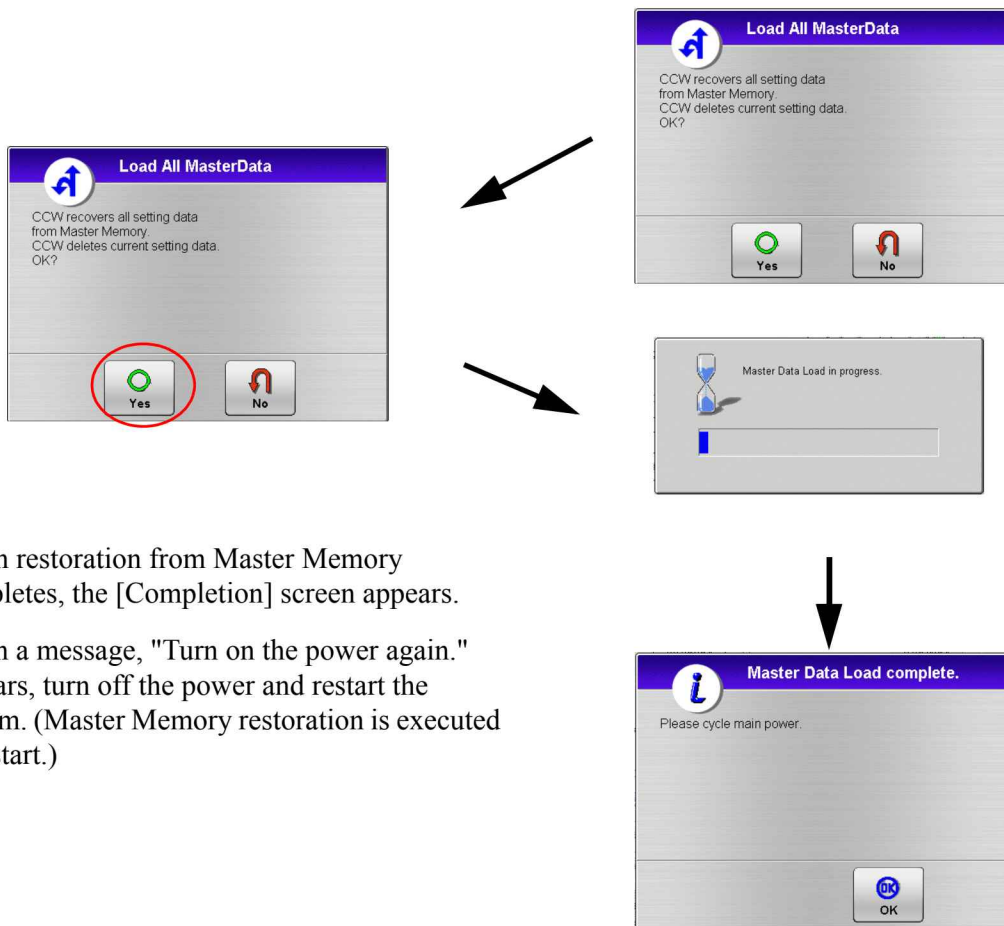
- A confirmation screen appears for restoration from Master Memory. To perform restoration, touch the



key.

To cancel restoration, touch the  key, and the screen returns to the previous screen.





3. When restoration from Master Memory completes, the [Completion] screen appears.

When a message, "Turn on the power again." appears, turn off the power and restart the system. (Master Memory restoration is executed at restart.)

Saving data to Master Memory

NOTE

- Saving the setting data to Master Memory after performing "Machine set" or "Preset" will overwrite the setting data of Master Memory, which means the previous setting data is deleted.
- When "Machine set" or "Preset" is additionally changed after delivery, ensure to save Master Memory to the CF memory in DMU board by [Save Master Memory].

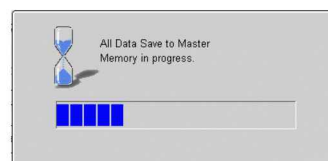
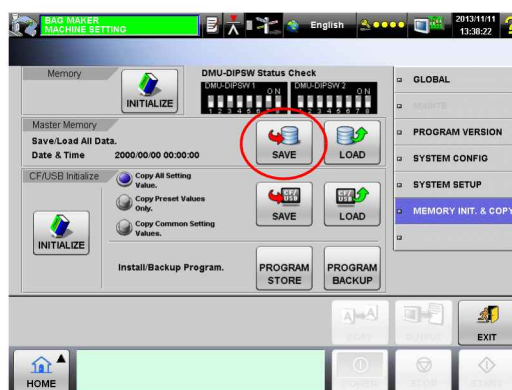
1. To save all the setting data to Master Memory, touch

the  key.

2. A confirmation screen appears for saving data to Master Memory. To perform saving, touch the

 key.

To cancel saving, touch the  key to return to the previous screen.



3. When saving to Master Memory is completed, the [Completion] screen appears.

Touching the  key saves all the setting value to Master Memory, and the screen returns to the previous one.



5.1.5.4 CF/USB Initialization

The CF/USB initialization function enables to save "All Setting Value", "Machine Set Value", "Preset Value" after delivery to the USB memory, and restore them from the USB memory to the CF memory in DMU board.

TIP

- After delivery, user can temporarily save "All Setting Value" into the USB memory in replacing the CF memory in DMU board. After replacement, the data can be restored to the CF memory in DMU board from the USB memory.

NOTE

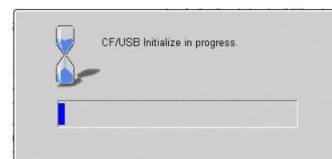
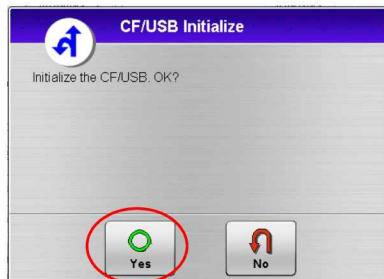
- When the machines have same specification and the same model, "Machine Set Value" and "Preset Value" can be saved from one machine to the USB memory, restored in or copied to another machine from the USB memory.
(When the software versions are different, this operation is not always successful.)

- To perform USB initialization, touch the  key.

A confirmation screen appears for USB initialization.

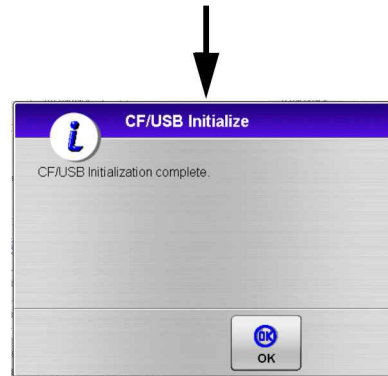
- To perform initialization, touch the  key.

To cancel initialization, touch the  key to return to the previous screen.



3. When the USB initialization completes, a message "Turn ON the power again." appears.

Touch the  key, turn off the power, and restart the system.
(Memory initialization is executed at restart.)

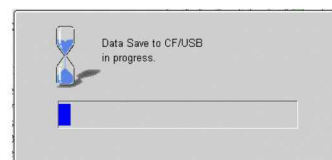


4. To save "All Setting Data" to USB memory, touch the [Copy All Setting Data] key, and touch the



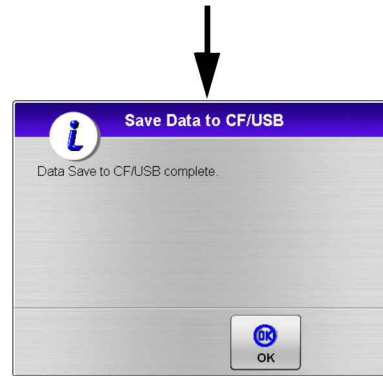
5. A confirmation screen appears for saving to the USB memory. To perform saving, touch the  key.

To cancel saving, touch the  key to return to the previous screen.



6. When saving [All Setting Value] to the USB memory completes, the [Completion] screen appears.

Touching the  key returns to the previous screen.

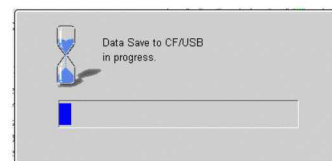


7. To save "Preset Value" to the USB memory, touch the [Copy Preset Value] key, and touch the



8. A confirmation screen appears for saving data to the USB memory. To perform saving, touch the  key.

To cancel saving, touch the  key to return to the previous screen.



9. When saving [Preset Value] to the USB memory completes, the [Completion] screen appears.

Touching the  key returns to the previous screen.

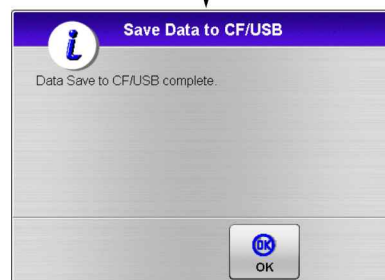
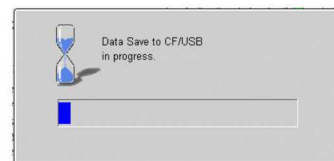


10. To save "Machine Setting Value" to USB memory, touch the [Copy Machine Setting Value] button, and

touch the  key.

11. A confirmation screen appears for saving data to the USB memory. To perform saving, touch the  key.

To cancel saving, touch the  key to return to the previous screen.



12. When saving [Machine Setting Value] to the USB memory completes, the [Completion] screen appears.

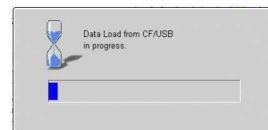
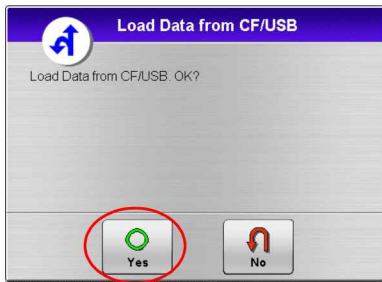
Touching the  key returns to the previous screen.

13. To restore "All Set Value" from the USB memory, touch the [Copy All Setting Value] button, and touch the  key.



14. A confirmation screen appears for restoring the data from the USB memory. To perform restoration, touch the  key.

To cancel restoration, touch the  key to return to the previous screen.



15. When restoration of "All Setting Value" from the USB memory completes, the [Completion] screen appears.

Touching the  key returns to the previous screen.



16. To restore "Preset Value" from the USB memory, touch the [Copy Preset Value] button, and touch the

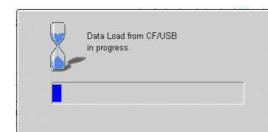
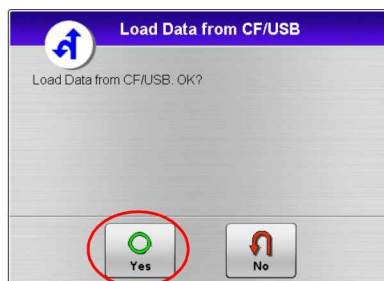


key.

17. A confirmation screen appears for restoring the data from the USB memory. To perform restoration,

touch the  key.

To cancel restoration, touch the  key to return to the previous screen.

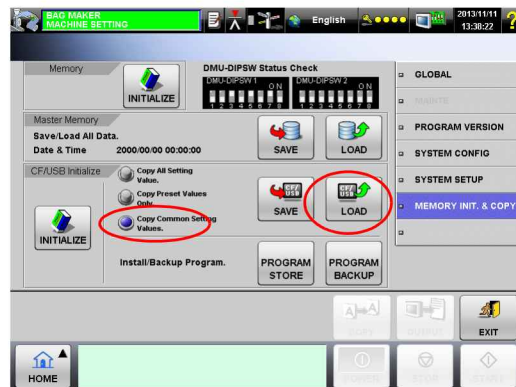


- 18 When restoration of "Preset Value" from the USB memory completes, the [Completion] screen appears.

Touching the  key returns to the previous screen.

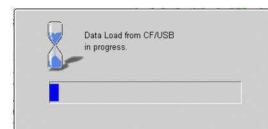
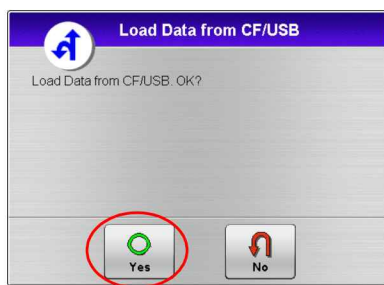


19. To restore "Machine Setting Value" from the USB memory, touch the [Copy Machine Setting Value] button, and touch the  key.



20. A confirmation screen appears for restoring the data from the USB memory. To perform restoration, touch the  key.

To cancel restoration, touch the  key to return to the previous screen.



21. When restoration of "Machine Setting Value" from the USB memory completes, the [Completion] screen appears.

Touching the  key returns to the previous screen.



5.2 DMU Board and MCU/SCU Board Software Installation Procedure

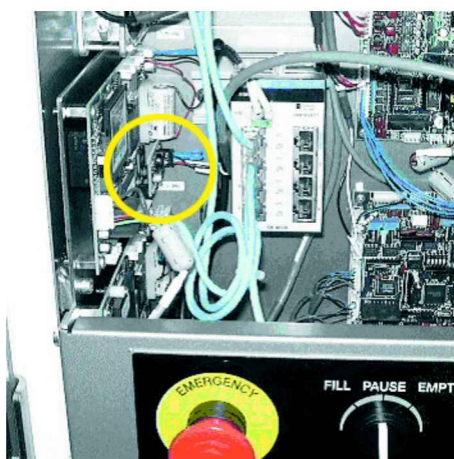
5.2.1 INTRODUCTION

This is a procedure of software installation for ATLAS-204/234 main body which has flash memory type micro processing unit (MITSUBISHI M16C).

Referential program and boards are as below.

Programme	Board	Micro Processing Unit	Installation Connector
MCU1	P-5547*	IC20	XJ23
MCU2		IC23	XJ21
SCU1	P-5548*	IC19	XJ19
SCU2	P-5549*	IC10	XJ23

Mount communication board (P-5475*) to each installation connector and then connect the cable coming from BM-DMU board (P-5562-4, yellow circled in below) to the connector (J791) on this communication board.



WARNING

- Do not connect the installation cable from BM-DMU board directly to installation connectors on each board without mounting the communication board (P-5475*).

When ATLAS-204/234 is dispatched from the victory, communication board (P-5475*) is mounted on the installation connector of SCU1 (XJ19 of P-5548*). Mount the communication board to a relevant installation connector for program to be installed.

Put back the communication board (P-5475*) to the original position (XJ19 of P-5548*) after program installation is completed.

5.2.2 Main Body Software Installation

If power supply is turned on with the corresponding harness wiring and dipswitch settings on each board, software writing starts automatically.

Software writing status is displayed with LED (H12-H15) on DMU board.

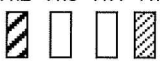
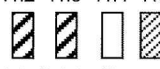
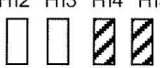
1. Downloading from DMU H12 H13 H14 H15  H12, H15: flashing alternatively (every second) H13, H14: light off <u>DO NOT TURN OFF POWER</u>	2. Uploading from DMU H12 H13 H14 H15  H14: light off H12, H13 and H15 : flashing alternatively (every second) <u>DO NOT TURN OFF POWER</u>	3. Complete writing DMU software H12 H13 H14 H15  H12, H13, H14 and H15 : flashing alternatively (every second)
4. Connection established, but ID unmatched (Error code = 109) H12 H13 H14 H15  Unmatched CPU dipswitch or faulty board? H12: light on, H13: light off H14, H15: flashing (every second)	5. Error when CF program is being read (Error code = 109) H12 H13 H14 H15  Needs to check CF program? H12: light on H13: light off H14, H15: flashing (every second)	6. Error when program is being downloaded (Error code = 109) H12 H13 H14 H15  Connection error of cable/board? H12: light off H13: light on H14, H15: flashing (every second)
7. Error when program is being uploaded or verified (Error code 109) H12 H13 H14 H15  Connection error of cable/board? H12, H13: light on H14, H15: flashing (every second)		

Fig. 5-1 Software writing status is displayed with LED (H12-H15) on DMU board.

When WEB-RCU boot up is made during DMU software writing procedure, the display showing software writing process is showed up.

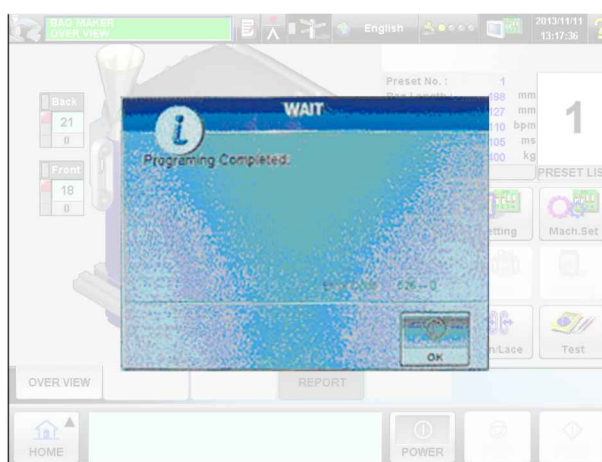
Press OK button at the right bottom of the message in order to turn off alarm buzzer.



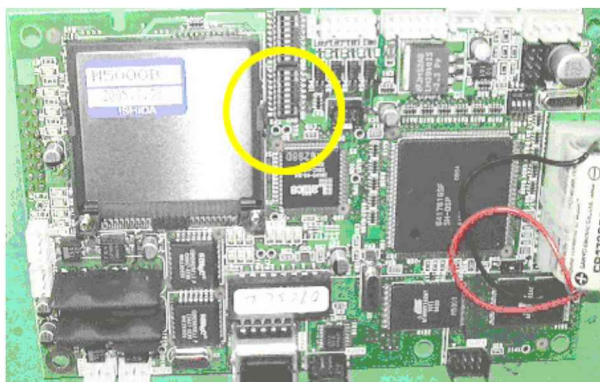
! WARNING

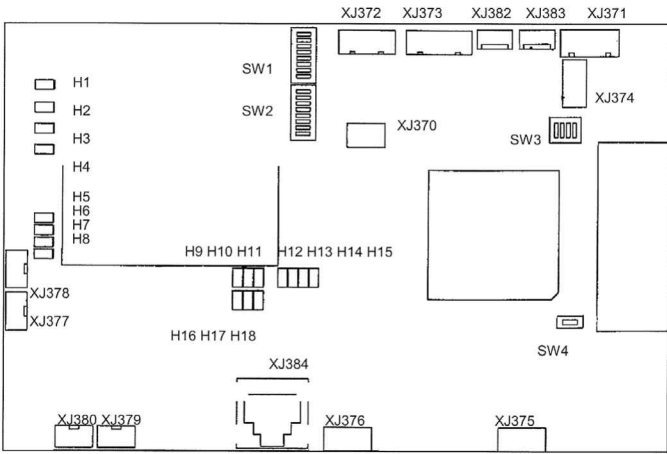
- Do not turn OFF machine power supply when the screen above with error code 525 - 0) is displayed.

When software writing completes, the following display appears.



5.2.3 Dip switch setting on DMU board





The dipswitch related to the main body software installation is SW2.

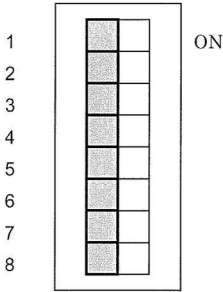
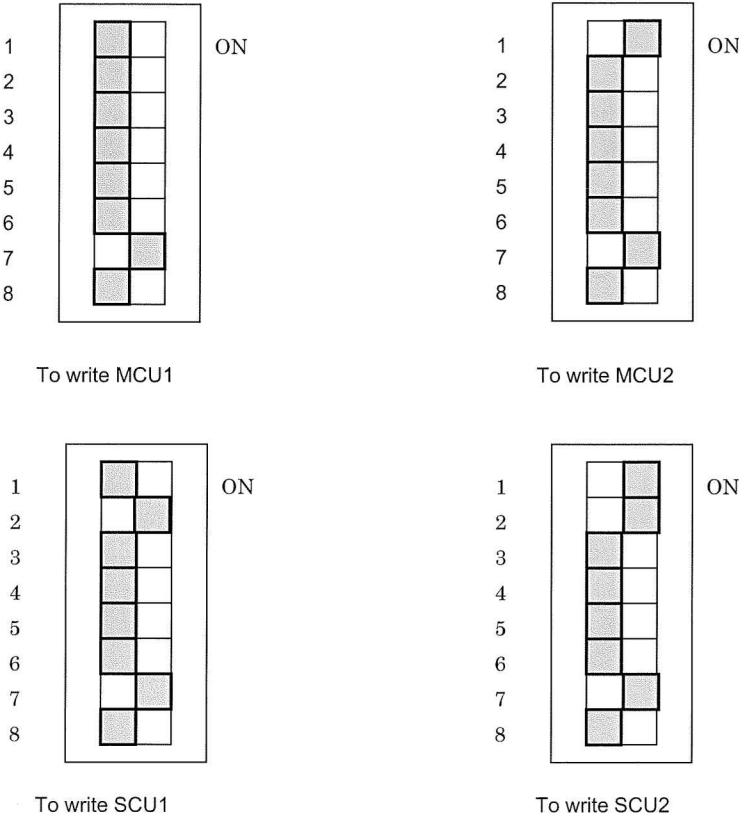


Fig. 5-2 On Ex-factory (Settings after software writing)



5.2.4 Setup for ATLAS

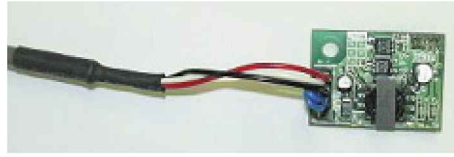
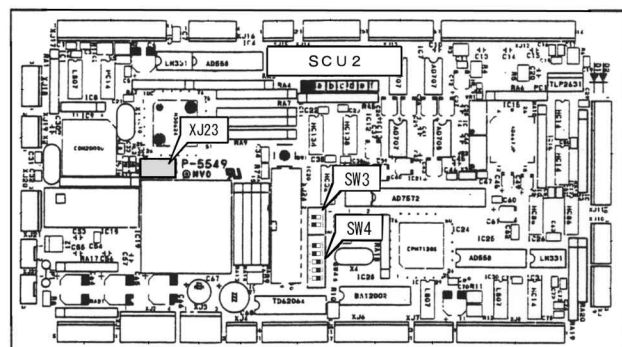
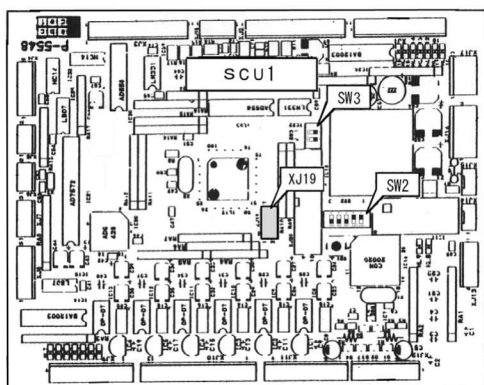
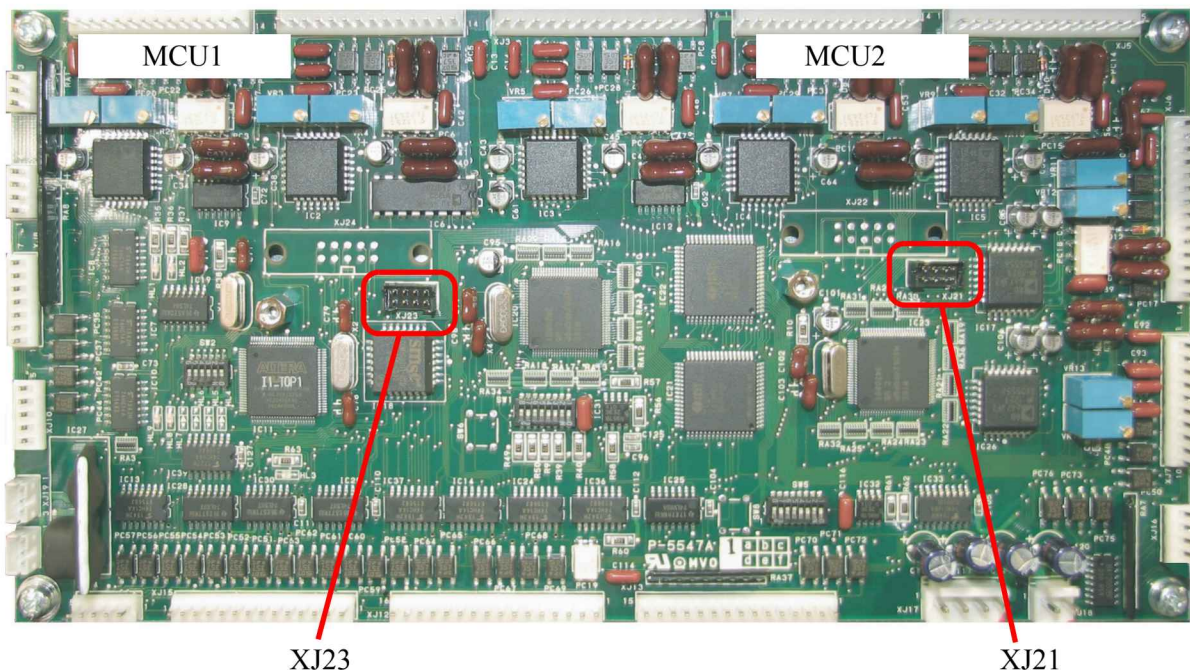


Fig. 5-3 Connection of communication board

5.2.4.1 Connection of communication board

After turning off power supply, connect the communication board with a cable (as in the photo above) to an installation connector for software to be installed.



5.2.4.2 Setting of dip switch

For software installation, setting change of dip switches is necessary. After turning off the machine power supply, set dip switches as below.

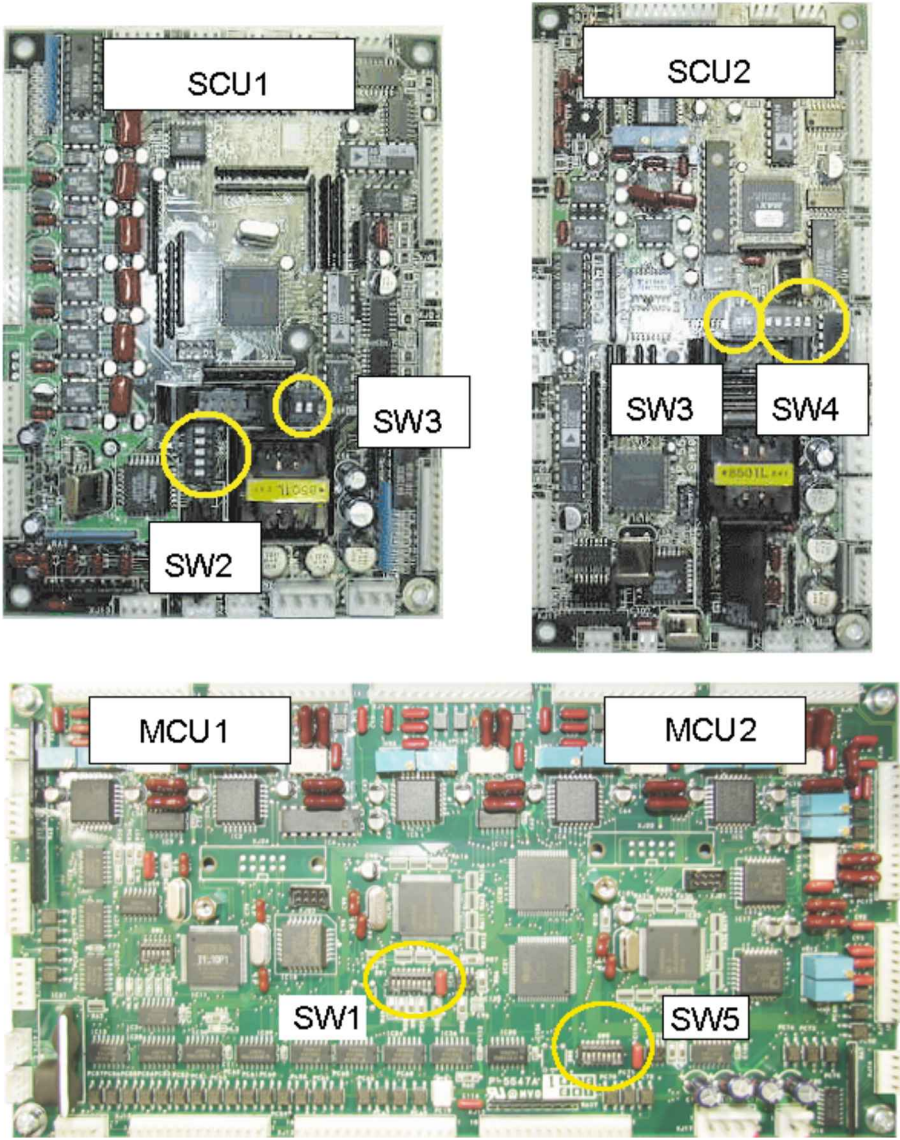


Table 5-1 Dip switch for MCU1 (P-5547*)

SW	On usual operation								On installation							
SW1	87654321 OFF ON OFF OFF ON ON OFF ON								87654321 OFF ON ON ON OFF OFF ON OFF							
	→ ←															

Table 5-2 Dip switch for MCU2 (P-5547*)

SW	On usual operation		On installation
SW5		→ ←	

Table 5-3 Dip switch for SCU1 (P-5548*)

SW	On usual operation		On installation
SW2		→ ←	
SW3		→ ←	

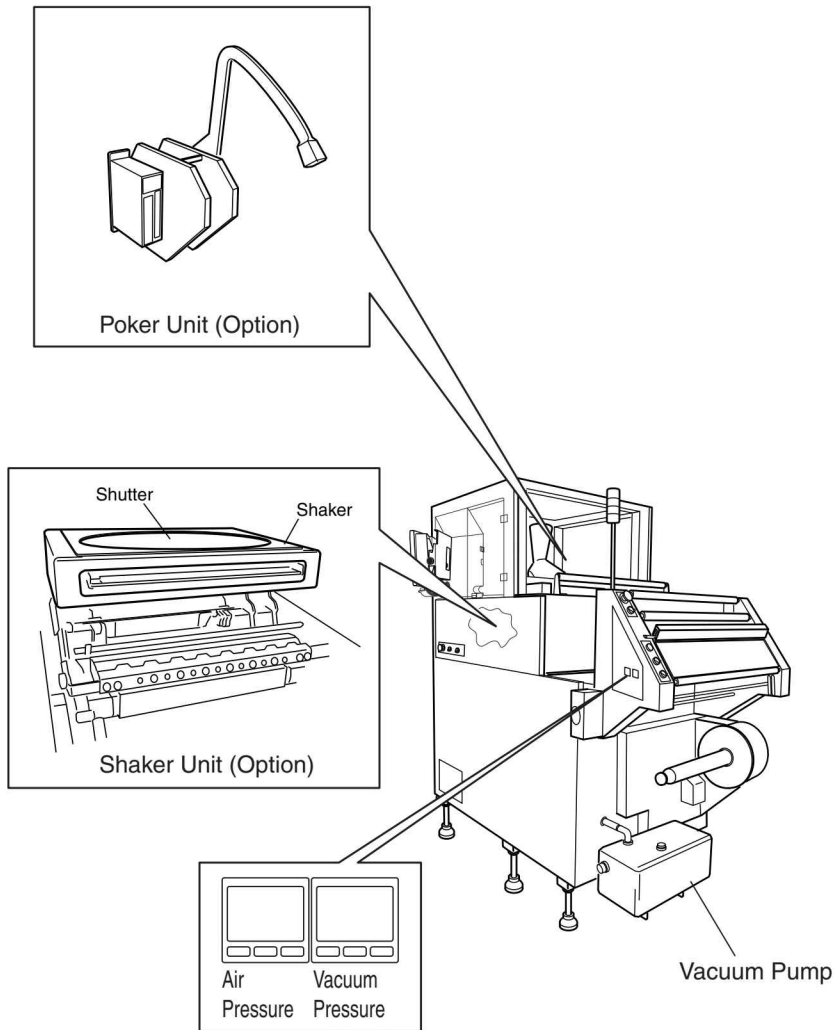
Table 5-4 Dip switch for SCU2 (P-5549*)

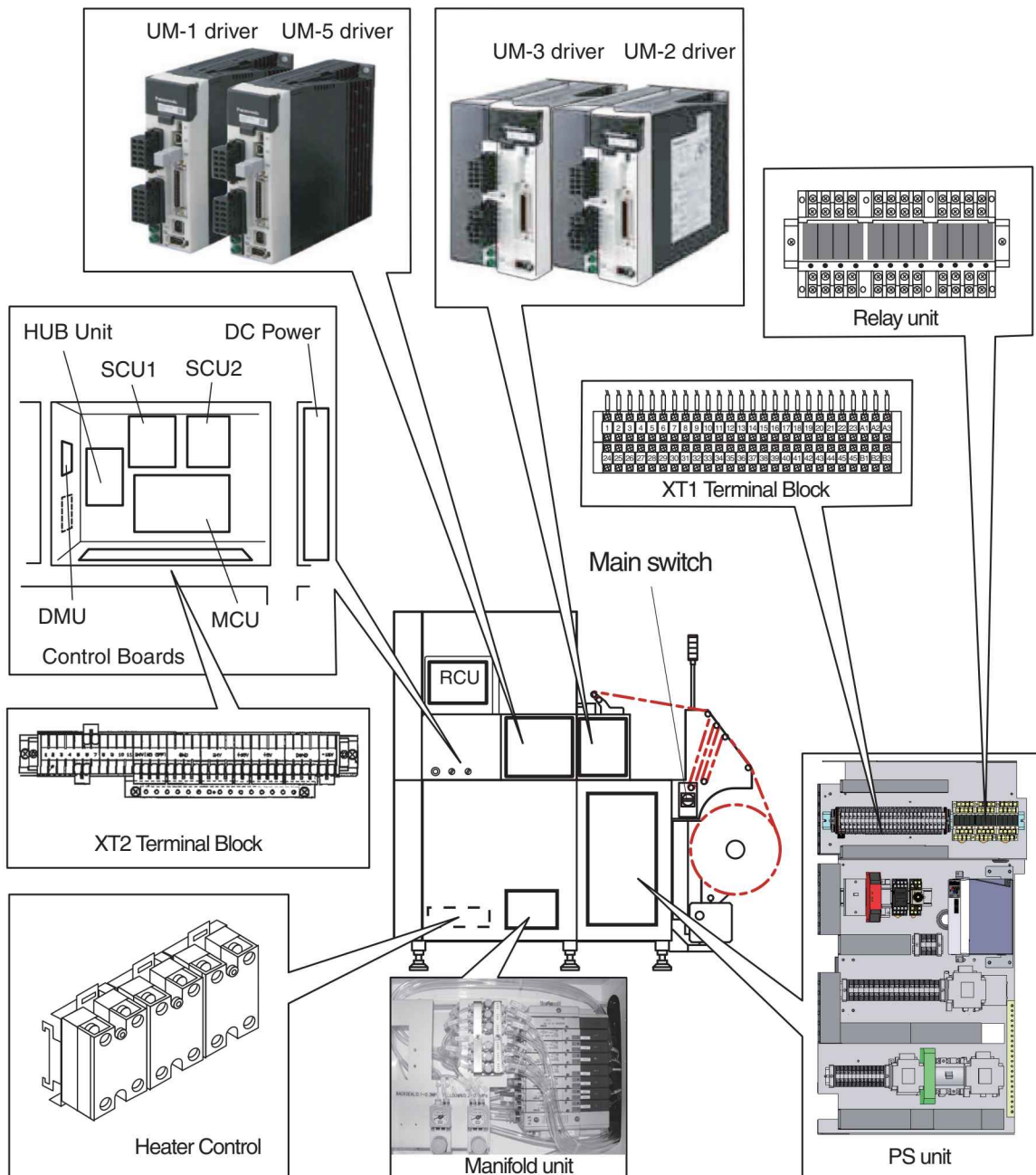
SW	On usual operation		On installation
SW4		→ ←	
SW3		→ ←	

Reset of dip switches

After software installation, reset the dip switches to the original status.

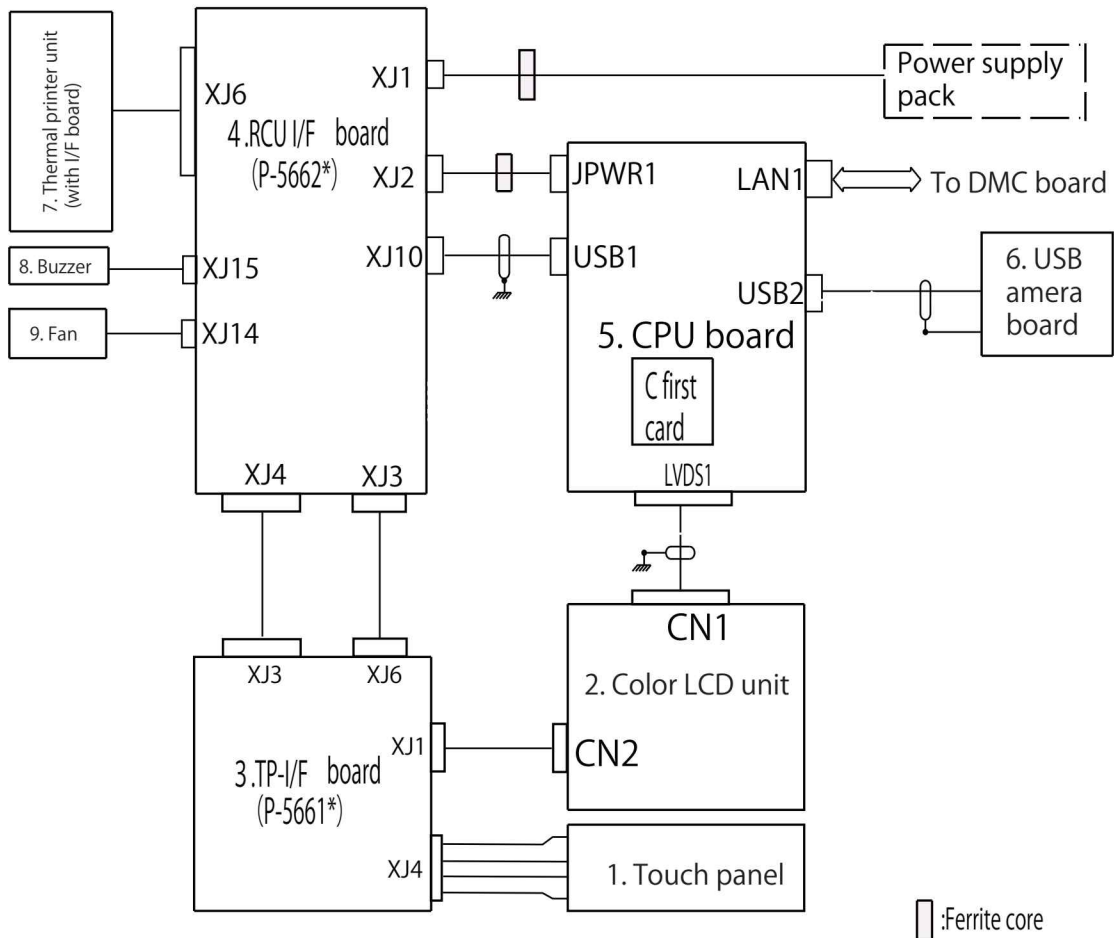
5.3 The locations of electrical boards and drivers





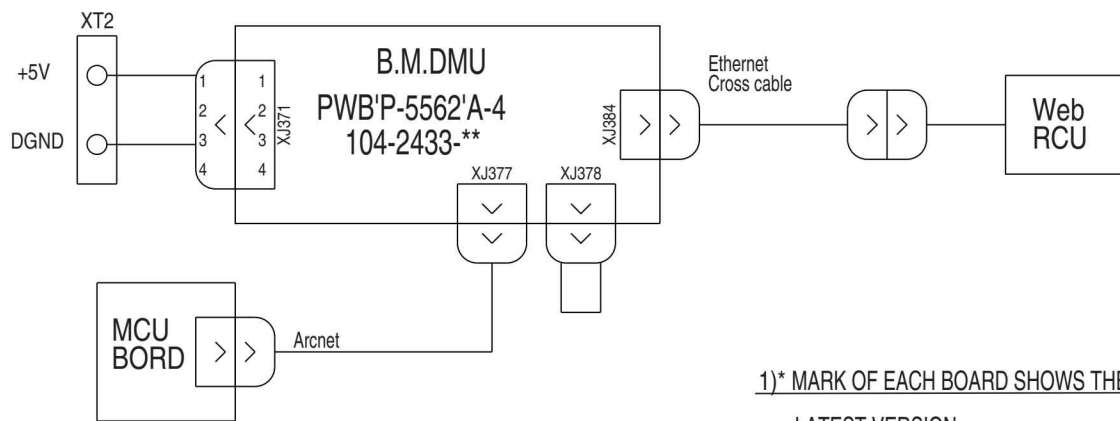
5.4 Parts of each boards

5.4.1 RCU Block diagram



Symbol	Description	Type no. / Notes	Maker
1	Touch panel	KBT-12.1C10R-FM	KB Eltron
2	Color LCD unit	AC121SA01	Mitsubishi
3	TP-I/F board (P-5661*)		
4	RCU-I/F board (P-5662*)		
5	CPU board	NPC-M0103	Omron
6	USB camera board		
7	Printer	SAM-1245-10K	SEIKO
8	Buzzer		
9	Fan		
10	USB camera switching board (P-5582*)	(option)	

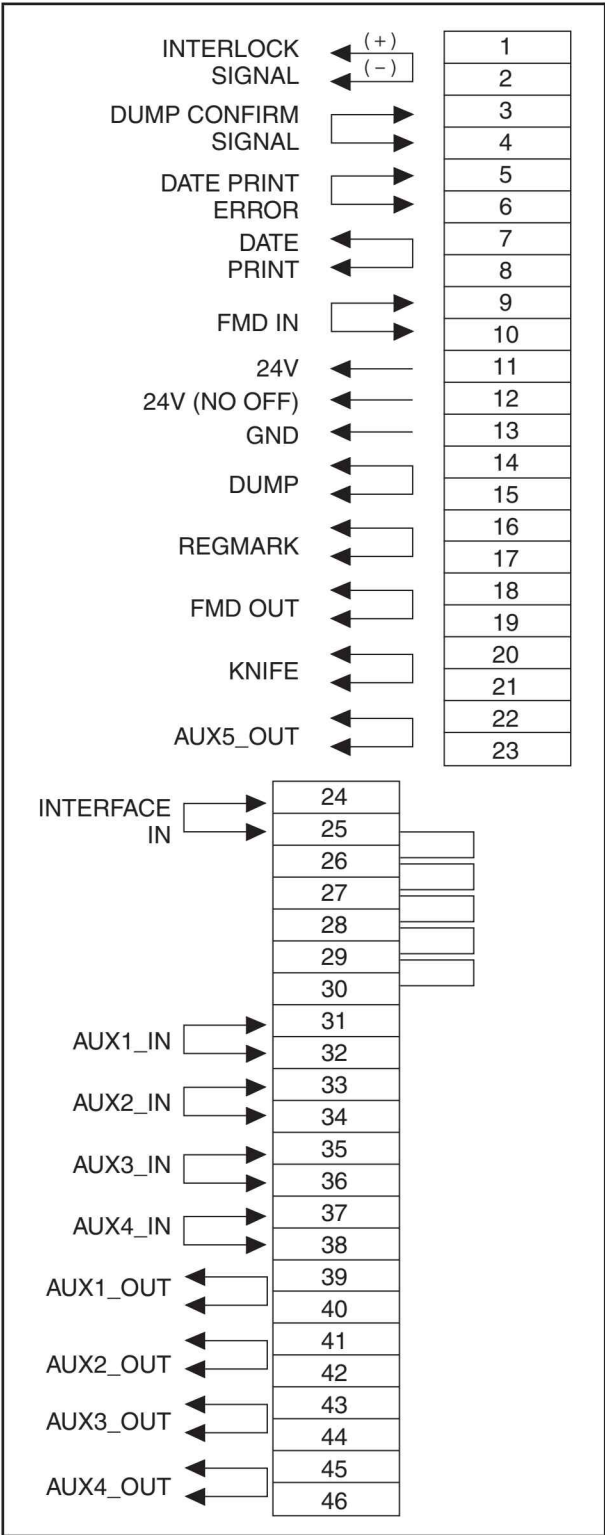
5.4.2 Block diagram



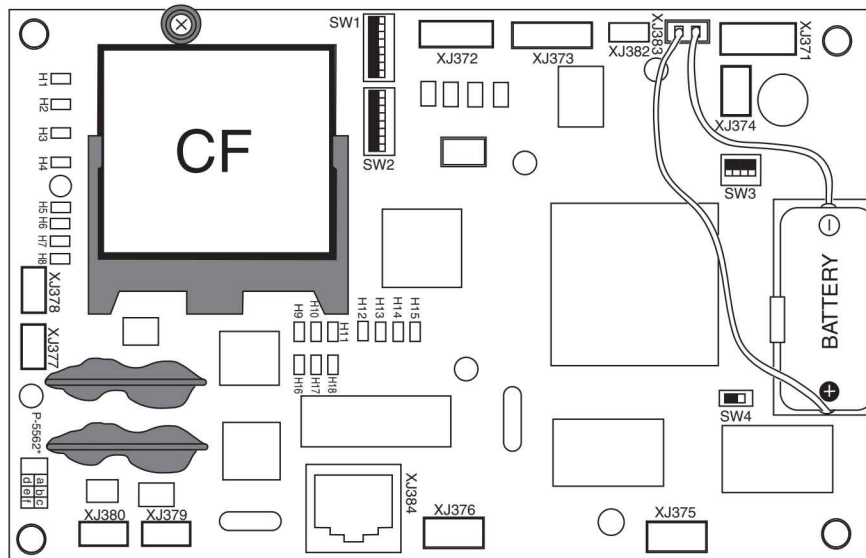
1)* MARK OF EACH BOARD SHOWS THE
LATEST VERSION.

WIRING DIAGRAM:: 111-0669-01

5.4.3 Wiring of XT1 Block



5.5 DMU BOARD (P-5562*)



Function of board

1. Data processing
2. Memory of Total and Parameter

Setting board DIP switches

(1) Setting off SW3

DIP SW	Function	Default at shipment
SW3-1	ON: Boot loader mode OFF: Application execution mode	OFF
SW3-2	Fix at OFF; do not turn ON.	OFF
SW3-3	ON: Flash memory write is inhibited. OFF: Flash memory write is enabled.	OFF
SW3-4	Fix at OFF; do not turn ON.	OFF

(4) Setting SW4 (Normally OFF)

Writing flash memory	OFF: Writing prohibited (available only with RS232 mounted)
	ON: Writing available

Battery

Battery in use: Lithium battery (Storage battery) CR17335SE-H SANYO

When replacing battery, connect new battery to XJ41. If an old battery is already connected, connect new battery to XJ40. Then remove old battery from another connector.

**CAUTION**

If the battery is replaced with an incorrect one, it may result in a board malfunction. When replacing a battery, make sure that it is the same model or an equivalent.

Dispose of used batteries after replacement.

Connector classification

Connector No.		Item
XJ370	(6P)	JTAG for CPLD (IC8)
XJ371	(4P)	DMU power supply
XJ372	(10P)	PI input CH1
XJ373	(12P)	PO output CH1
XJ374	(8P)	JTAG for CPU
XJ375	(8P)	Serial interface CH0
XJ376	(8P)	Serial interface CH1
XJ377	(3P)	Arcnet CH1
XJ378	(3P)	Arcnet CH1
XJ379	(3P)	Arcnet CH0
XJ380	(3P)	Arcnet CH0
XJ381		Compact flash
XJ382	(2P)	Battery
XJ383	(2P)	Battery
XJ384		ETHERNET connector
XJ385	(20P)	Interface for DMU
XJ386	(10P)	Power supply for stack

Connector function in details

(1) XJ370: JTAG for CPLD (IC8)

Connector No.	Terminal No.	Signal
XJ370 JTAG for CPLD (IC8)	1	+3.3V
	2	TMS
	3	TDO
	4	TDI
	5	TCK
	6	GND

(2) XJ371: DMU power supply

Connector No.	Terminal No.	Signal
XJ371 DMU power supply	1	+5V DC
	2	+5V DC
	3	GND
	4	GND

(3) XJ372: PI input CH1

Connector No.	Terminal No.	Signal
XJ372 PI input CH1	1	+ common
	2	PINT 0
	3	PINT 1
	4	PINT 2
	5	PINT 3
	6	PINT 4
	7	PINT 5
	8	PINT 6
	9	PINT 7
	10	N.C.

(4) XJ373: PO input CH1

Connector No.	Terminal No.	Signal
XJ373 PO output CH1	1	Data 0
	2	Data 1
	3	Data 2
	4	Data 3
	5	Data 4
	6	Data 5
	7	Data 6
	8	N.C.
	9	Data 7
	10	+ 5V
	11	GND
	12	N.C.

(5) XJ374: JTAG for CPU

Connector No.	Terminal No.	Signal
XJ374 JTAG for CPU	1	TCK
	2	TRST
	3	TDO
	4	GND
	5	TMS
	6	TDI
	7	RES
	8	GND

(6) XJ375: Serial interface CH0

Connector No.	Terminal No.	Signal
XJ375 Serial interface CH0	1	+5V
	2	TXD
	3	RXD
	4	GND
	5	RTS
	6	CTS
	7	DSR
	8	DTR

(7) XJ376: Serial interface CH1

Connector No.	Terminal No.	Signal
XJ376 Serial interface CH1	1	+5V
	2	TXD
	3	RXD
	4	GND
	5	RTS
	6	CTS
	7	DSR
	8	DTR

(8) XJ377/ XJ378: Arcnet CH1

Connector No.	Terminal No.	Signal
XJ377, XJ378 Arcnet CH1	1	Signal (+)
	2	Signal (-)
	3	F.G.

(9) XJ379/XJ380: Arcnet CH0

Connector No.	Terminal No.	Signal
XJ379, XJ380 Arcnet CH0	1	Signal (+)
	2	Signal (-)
	3	F.G.

(10) XJ385: Interface for DMU

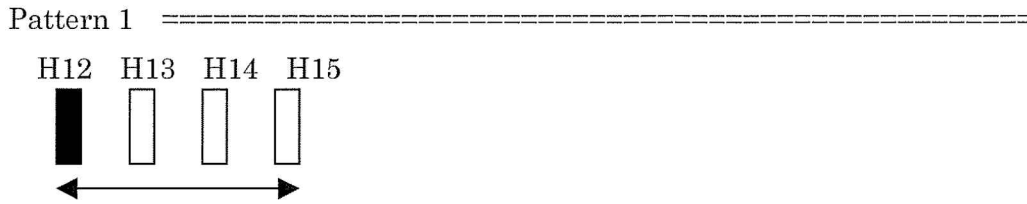
Connector No.	Terminal No.	Signal
XJ385 Interface for DMU	1	Data 0
	2	Data 1
	3	Data 2
	4	Data 3
	5	Data 4
	6	Data 5
	7	Data 6
	8	Data 7
	9	Address 0
	10	Address 1
	11	Address 2
	12	Address 3
	13	R/W direction input
	14	ENB input
	15	RESET input
	16	Interruption output
	17	Serial input signal (CH0)
	18	Serial output signal (CH0)
	19	+ 5V
	20	GND

(11) XJ386: Power supply for stack

Connector No.	Terminal No.	Signal
XJ386 Power supply for stack	1	+ 5V
	2	+ 5V
	3	+ 5V
	4	+ 5V
	5	GND
	6	GND
	7	GND
	8	GND
	9	GND
	10	GND

Function of LED for monitor

LED	Functional description	LED display status
H1	PINT2	Turns off in an unconnected state.
H2	PINT3	Turns off in an unconnected state.
H3	PINT0	Turns off in an unconnected state.
H4	PINT1	Turns off in an unconnected state.
H5	Arcnet CH1 Reception signal	Lights brightly.
H6	Arcnet CH1 Transmission signal	Lights poorly.
H7	Arcnet CH0 Reception signal	Lights brightly with an instant turn off at intervals in an unconnected state.
H8	Arcnet CH0 Transmission signal	Lights poorly with an instant turn off at intervals in an unconnected state.
H12	Indicates the boot loader status by combination of 12 to 15.	(*) Flashes with normal DMU software operation.
H13		(*) Flashes with receiving operation of main boat telegram.
H14		(*) Flashes with telegram receiving of MELSE Ethernet.
H15		(*) Flashes with receiving operation of MELSEC232 telegram.
H9	ETHER data reception	Flashes.
H10	ETHER link	Lights.
H11	ETHER all double communication	Lights.
H16	ETHER link with 100MBPS	Lights.
H17	ETHER data transmission	Flashes.
H18	ETHER collision occurred	Turns off when there is no collision.

Pattern 1

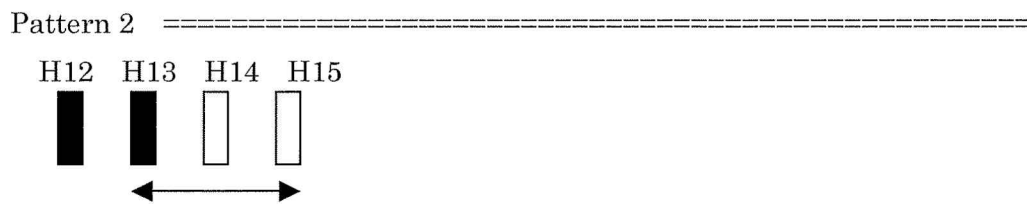
H12, H13, H14, and H15 light one after another in sequence:

The bootloader starts.

When SW3-1 is ON, restarts with turning OFF SW3-1.

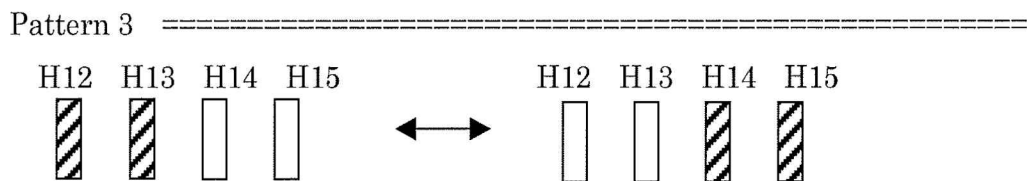
When SW3-1 is OFF, the flash memory has no application software and the CF card is not set. (When the CF card without INF files is set, the pattern is 5.)

When SW3-1 is OFF and only H12 flashes as shown in pattern 4 after a few seconds, it is possible that either a DMU board for other than the packer may be installed or the board may malfunction.

Pattern 2

While H12 lights, H13, H14, and H15 light one after another in sequence:

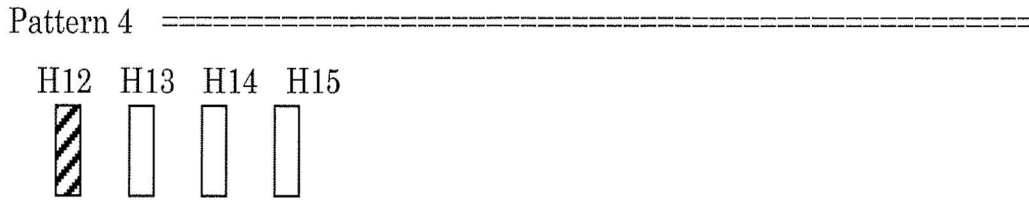
In the file creation process **(Do not turn off the main power during this process.)**

Pattern 3

H12 & H13 and H14 & H15 flash alternately:

In recovering the backup file to the root **(Do not turn off the main power during this process.)**

Pattern 4



Only H12 flashes:

In downloading and rewriting the DMU software (**Do not turn off the main power during this process.**),

or

in downloading the other software (after the transition from the pattern 1).

However, when this condition continues after several tens of minutes have passed, it is possible that the system was set to pattern 1 immediately after power on and transferred to this status. Turn off the main power and turn it on again, then immediately check the LED status.

Pattern 5



All of H12, H13, H14, and H15 flash together:

or

Commanding SRAM initialization (Drive Mount error: **Error code = 1019**).

When a broken CF card or a CF card without INF file is set into the socket, the system becomes this status even SRAM operates normally. Turns off the power before turning off SW2-R, then turn on the power again with the CF card removed to confirm the status.

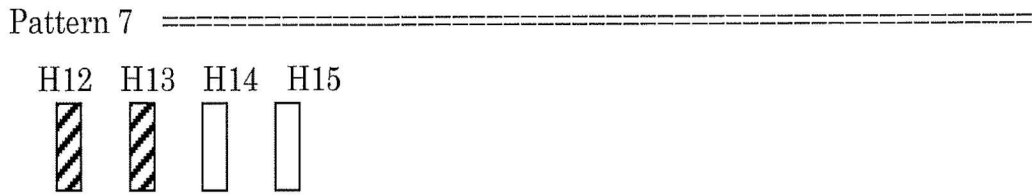
Pattern 6



H12, H13, and H14 flash together:

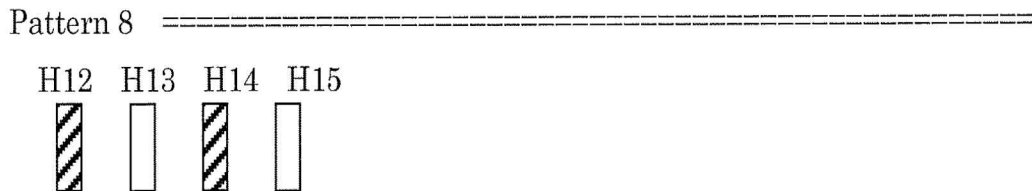
Commanding initialization of SRAM disk (Scan Disk error: **Error code = 1109**).

When a broken CF card or a CF card without INF file is set into the socket, the system becomes this status even SRAM operates normally. Turns off the power before turning off SW2-R, then turn on the power again with the CF card removed to confirm the status.

Pattern 7

H12 and H13 flash together:

Commanding initialization of SRAM disk (Setup file error: **Error code =1027**).

Pattern 8

H12 and H14 flash together:

Commanding initialization of SRAM disk (Log file error: **Error code =1023**).

Turning OFF SW2-6 in the state of pattern 5 to 8 performs initialization of SRAM disk.

The error code is not set while the LED flashes. It is set when SW2-6 is OFF and after the initialization is performed.

Pattern 9

H12 and H15 flash together:

Assert is generated.

Saves the file name and the number of lines that have generated in ASSERT.LOG of SRAM disk.

When the FTP connection is not possible in this state, turn on the main power again and make the FTP connection to acquire ASSERT.LOG.

```
ftp 192.168.0.41
```

```
user
```

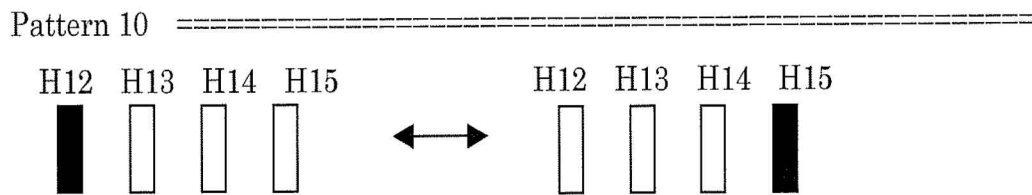
```
pass
```

```
cd c:
```

```
ls
```

```
bin
```

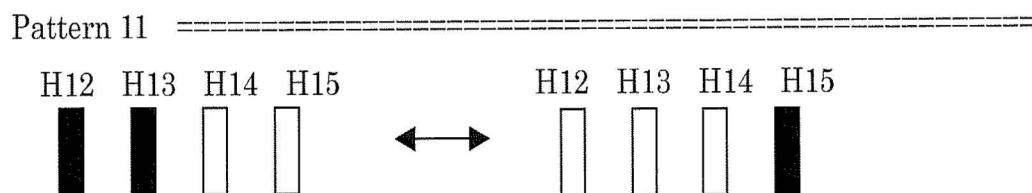
```
get assert.log
```


Pattern 10

While H13 & H14 turn off, H12 & H15 light alternately (at one second cycle):
In downloading the basic software from the DMU board: (**Error code = 525**).

NOTE

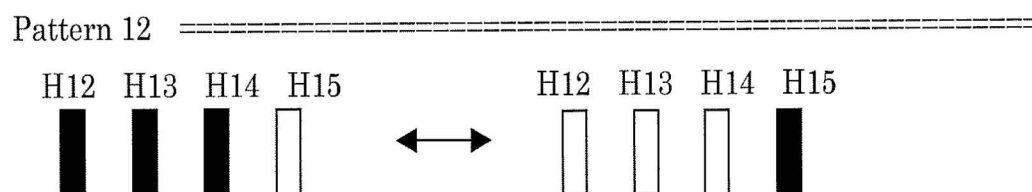
- Do not turn off the main power during this process.

Pattern 11

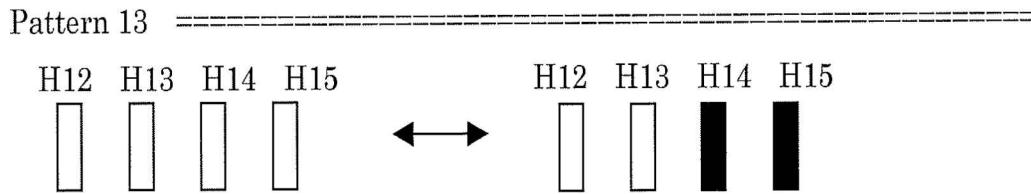
While H14 turns off, H12 & H13 and H15 light alternately (at one second cycle):
In uploading the basic software from the DMU board: (**Error code = 525**).

NOTE

- Do not turn off the main power during this process.

Pattern 12

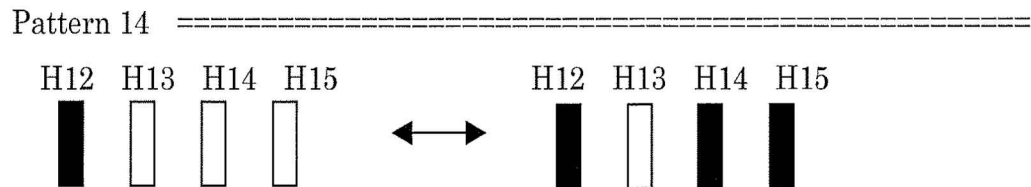
H12, H13, & H14 and H15 light alternately (at one second cycle):
Writing the basic software from the DMU board has been completed: (**Error code = 526**).

Pattern 13

While H12 & H13 turn off, H14 and H15 flash at one second cycle:

Although writing the basic software from the DMU board has completed the connection, the IDs were not identical (**Error code = 109**).

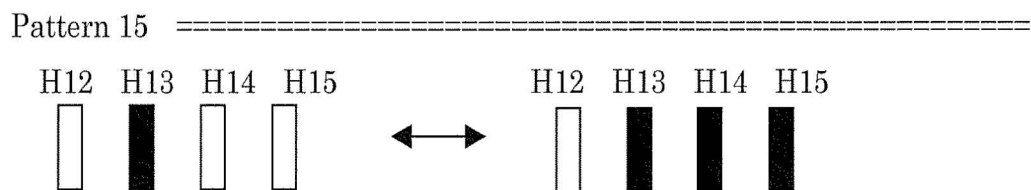
The cause may be due to a discrepancy between the connection destination and CPU selection dip switch or a connection failure.

Pattern 14

While H12 lights and H13 turns off, H14 and H15 flash at one second cycle:

In writing the basic software from DMU, an error occurs during the software reading on the CF card (**Error code = 109**).

Check the content of the CF card.

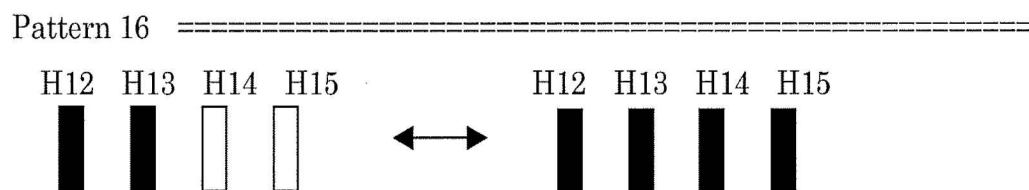
Pattern 15

While H12 turns off and H13 lights, H14 and H15 flash at one second cycle:

In writing the basic software from the DMU board, an error occurs during the download process (**Error code = 109**).

The cause may be a poor connection of the writing cable or the board.

Pattern 16

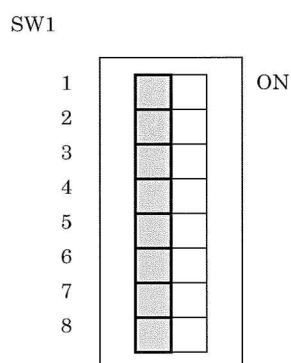


While H12 and H13 light, H14 and H15 flash at one second cycle:

In writing the basic software from the DMU board, an error occurs during the upload or verification process (**Error code = 109**).

The cause may be a poor connection of the writing cable or the board.

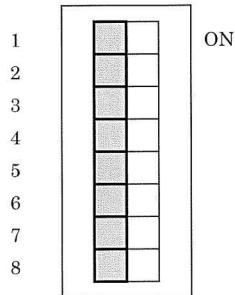
SW1



No.	Function	Normal
1	Reservation (F specification)	OFF
2	Reservation (model)	OFF
3	Right machine OFF, Left machine ON	OFF or ON
4	Modbus / TCP character string edian	OFF:DDE(big) / ON:OPC(little)
5	Soft stripping effective operation (specification:F)	OFF
6	Not used	OFF
7	Not used	OFF
8	Trial mode	OFF

SW2

SW2



No.	Function	Normal
1	Auto-basic software writing CPU selection#1	OFF
2	Auto-basic software writing CPU selection#2	OFF
3	Auto-basic software writing CPU selection#3	OFF or ON
4	Not used	OFF
5	Communication monitor output permission	OFF
6	Manual initialization of SRAM, CF CARD writing permission	OFF
7	Debug	OFF
8	BIOS log message transmission	OFF (Some machines have been set to ON at the factory.)

SW2-1;OFF, SW2-2;OFF, SW2-3:OFF\ \ \ MCU1
 SW2-1;ON, SW2-2;OFF, SW2-3:OFF\ \ \ MCU2
 SW2-1;OFF, SW2-2;ON, SW2-3:OFF\ \ \ SCU1
 SW2-1;ON, SW2-2;ON, SW2-3:OFF\ \ \ SCU2
 SW2-1;OFF, SW2-2;OFF, SW2-3:ON\ \ \ SCU3

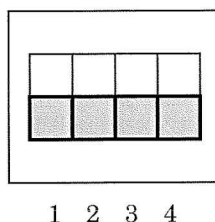
When SW2-8 is ON, the transfer rate of COM #1 (XJ375) is 57600 bps. When OFF, the rate is 38400 bps. **However, when SW2-8 is OFF, SW2-7 is ON, and SW2-5 is OFF, the system is set to the auto basic software writing mode**, which establishes the connection with the main CPU at 9600 bps and writes at 19200 bps after the completion of the connection. After the connection is established, the **Error code is 109**, displaying the progress with LED of H12 to H15. (Pattern 10 to pattern 16)

When SW2-5, SW2-7, SW2-8 are ON, the system automatically starts the scan disk of the CF card. It takes a few tens of seconds with 16 M. When the scan processing is unsuccessful including the case that the CF card is not inserted in the socket, the **Error code is 1051**. When any trouble is detected from the scan result, the **Error code is 1050**.

SW3

SW3

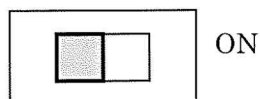
ON



No.	Function	Normal
1	Starting bootloader	OFF
2	Flash writing protection	OFF
3	Starting default IP address	OFF (192.169.0.41 when it is ON)
4	Starting JTAG-ICE connection	OFF (turn ON SW1-1 as well when it is ON)

SW4

SW4



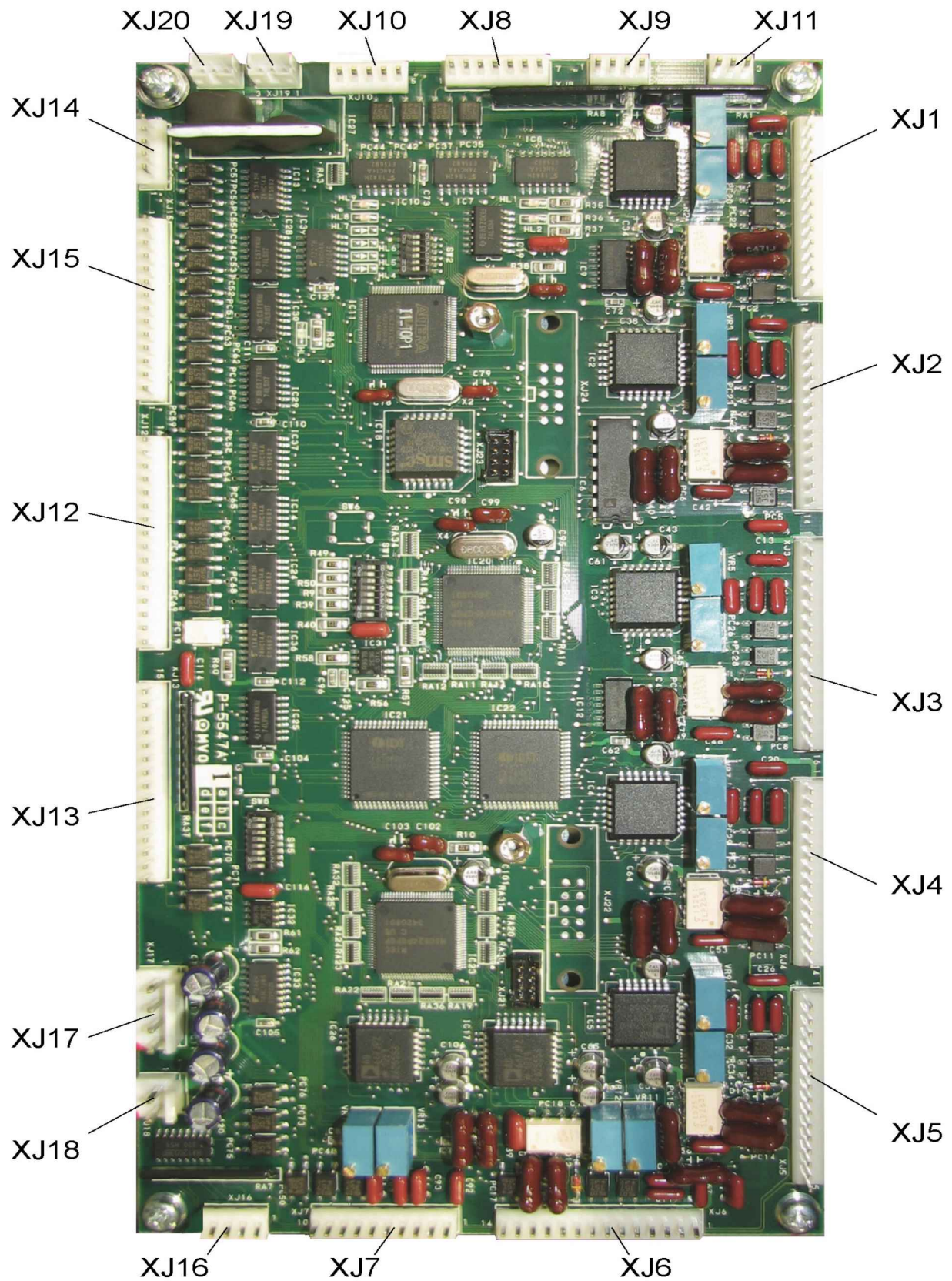
No.	Function	Normal
1	Write protection open	OFF

Connector

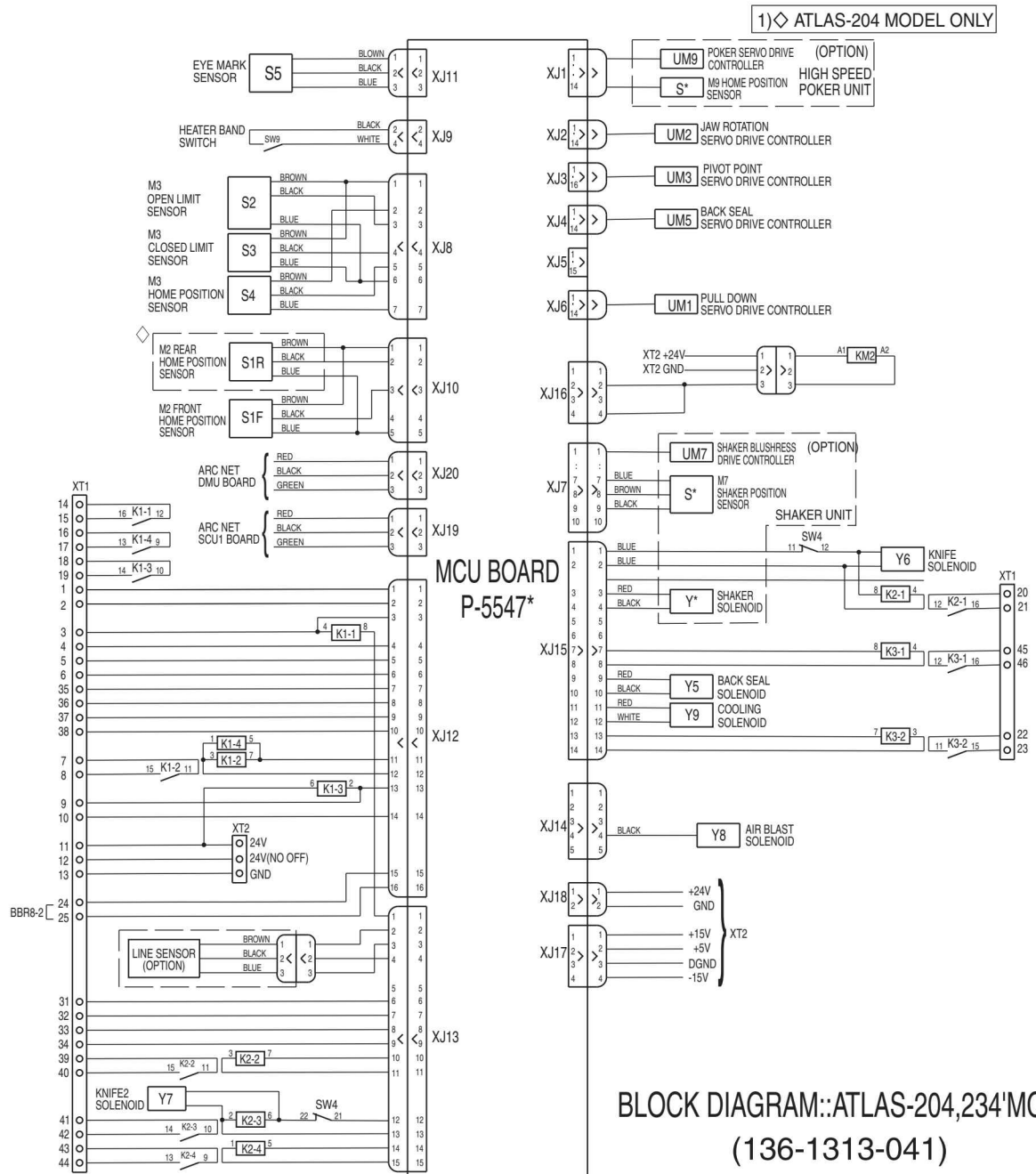
Symbol	Function
XJ370	Writing PLD
XJ372	8-IN
XJ373	8-OUT
XJ374	JTAG-ICE
XJ375	COM #1: Writing to Monitor or Main software.
XJ376	COM #2: MELSEC RS232C interface.
XJ377	ARCNET #1: Main unit (MCU, SCU1, SCU2) board communication.
XJ378	ARCNET #1: Normally terminating resistance.
XJ379	ARCNET #2: Optional board (2nd-DMU, MHIC) communication.
XJ380	ARCNET #3: Normally terminating resistance.

Symbol	Function
XJ381	CF card socket
XJ382	Battery #1
XJ383	Battery #2: Normally opened
XJ384	Ethernet
XJ385	Expansion bus

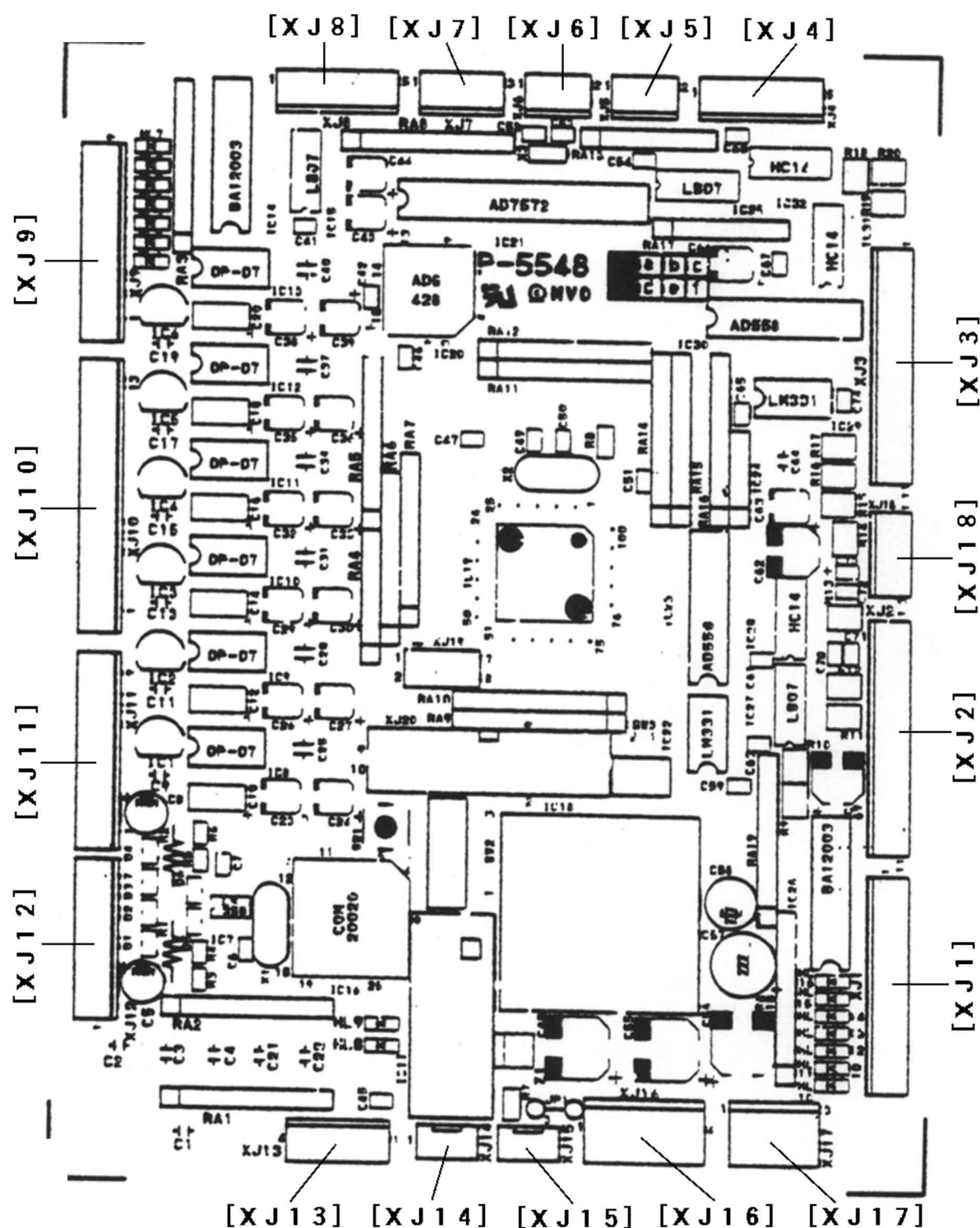
5.6 MCU board (P-5547*)



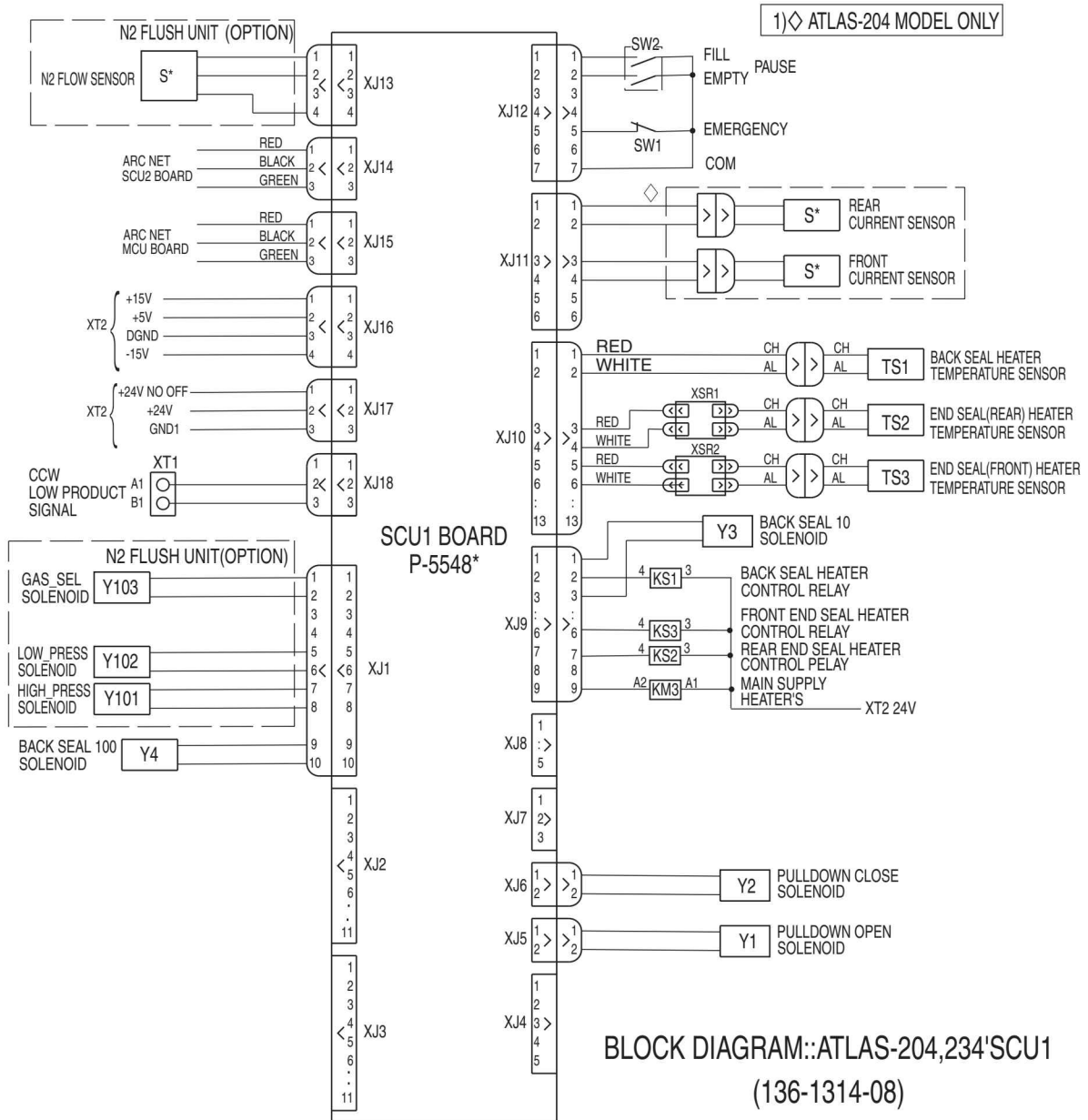
5.6.1 MCU board Block diagram



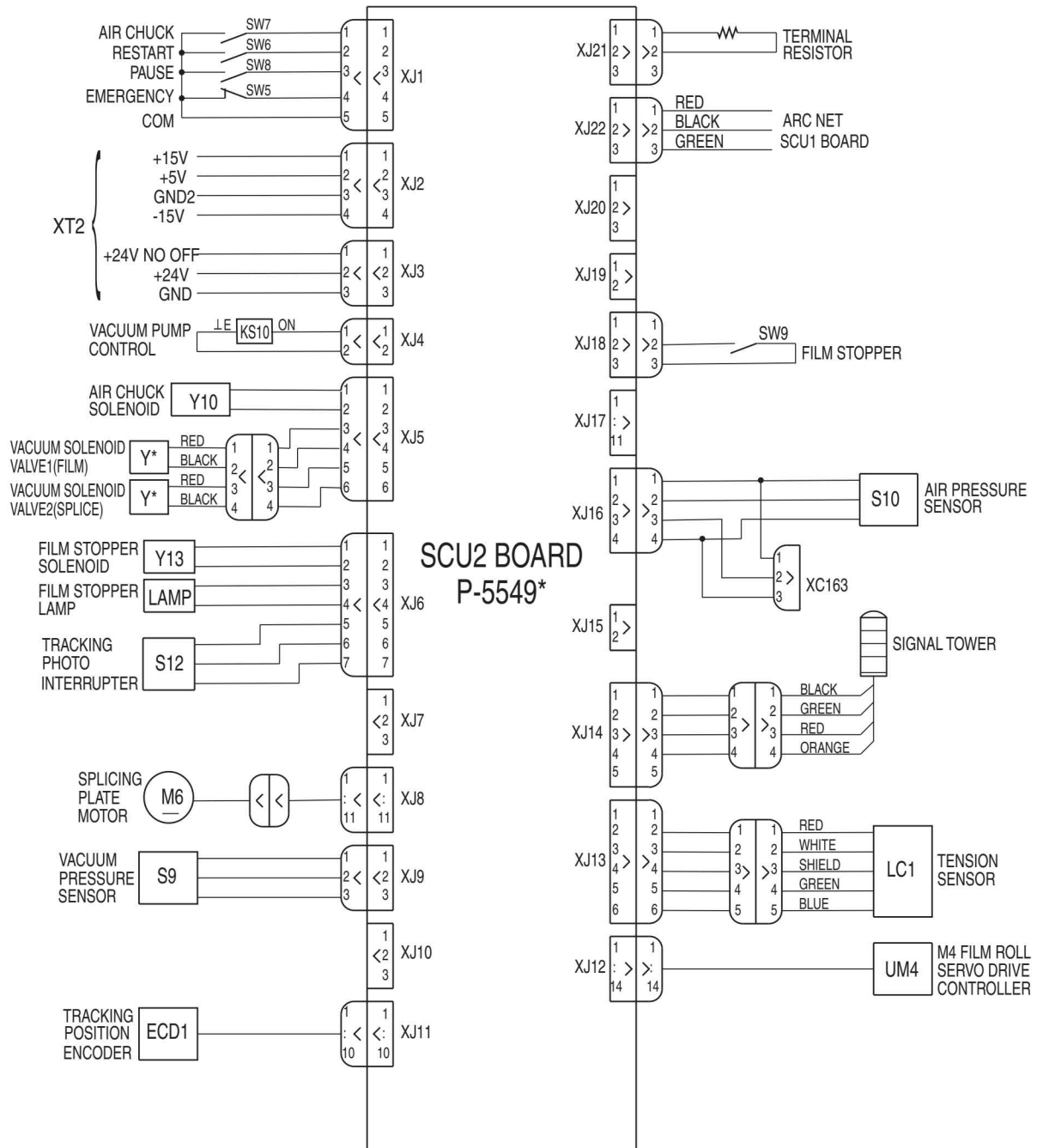
5.7 SCU1 board (P-5548*)



5.7.1 SCU1 board Block diagram

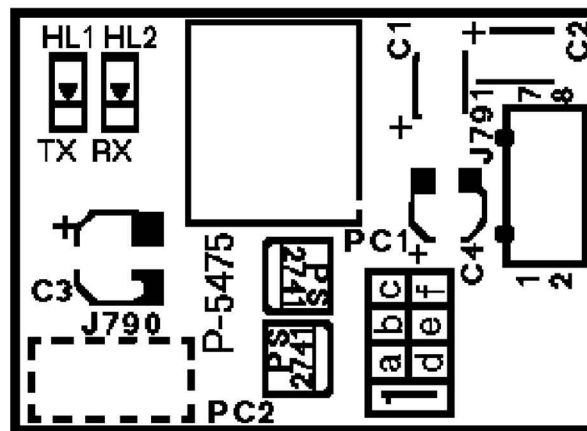


5.8.1 SCU2 board Block diagram

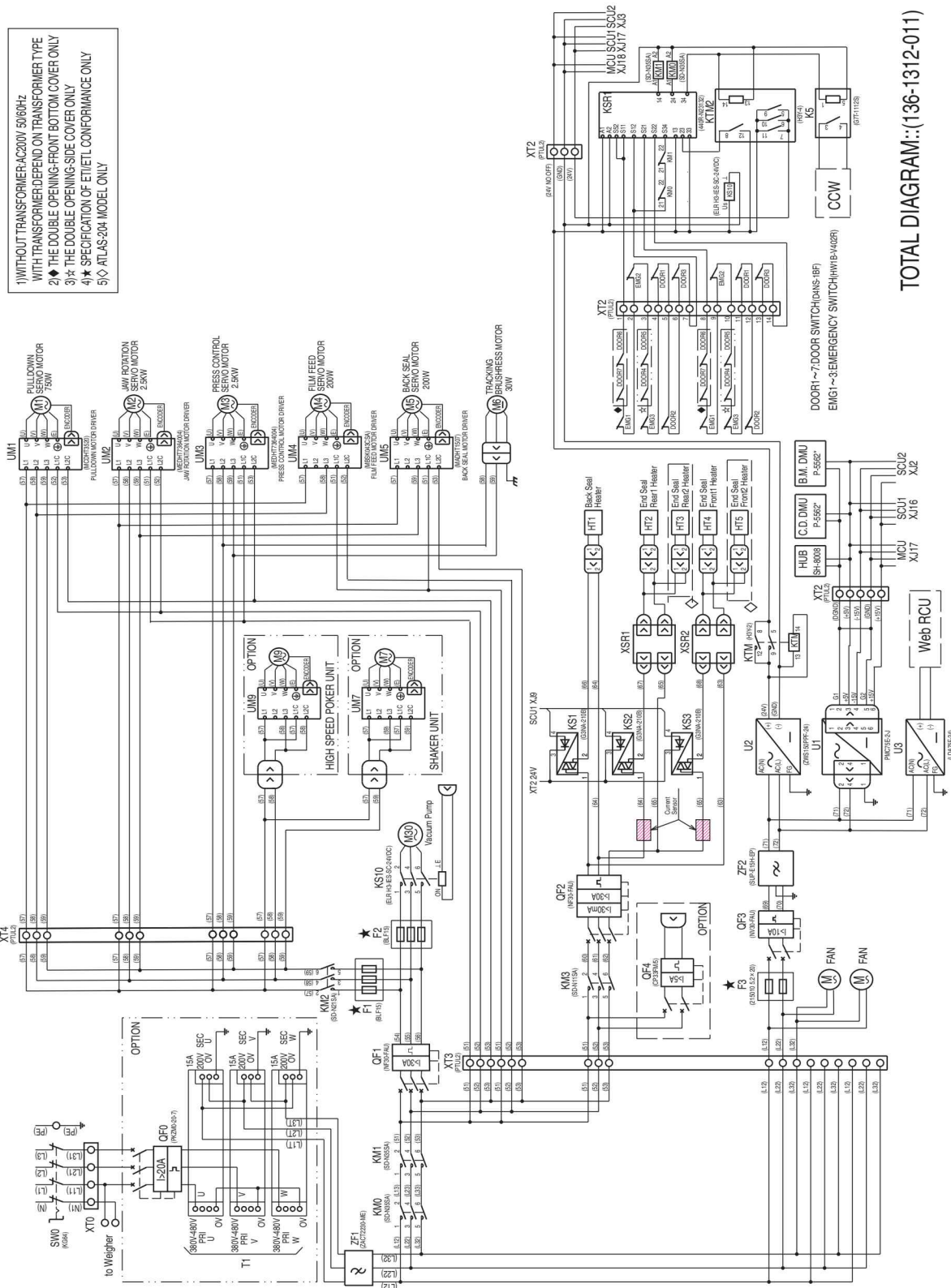


BLOCK DIAGRAM::ATLAS-204,234'SCU2
(136-1315-01)

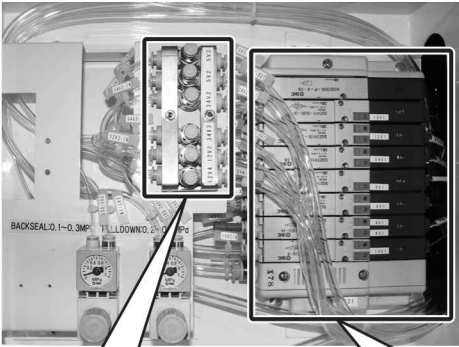
5.9 RS 232C Interface board (P-5475*)



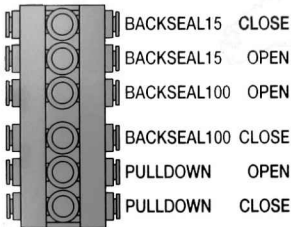
5.9.1 Total Diagram



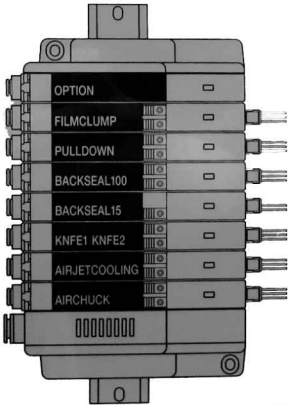
5.9.3 Manifold Arrangement



Air speed control



Layout drawing of the solenoid valves



ATLAS-204/234

5.10 Operation of the Motor Driver

- This driver have various parameters through which you can adjust/set the performance of functions. Please use these parameters so that you can operate the driver at the most appropriate condition.
- Among various functions of this driver are;
 - Monitoring functions of such as the number of reserved pulses of the deviation counter (position error), motor speed and generated torque.
 - Display of the status of the control input/output signals connected to the connector, CN I/F.
 - Display of the error factors and the record.
- There are 2 ways of operating the above functions as below;
 - Key operation and the display of the front panel or,
 - Computer display

5.10.1 AC Servo Driver

5.10.1.1 Key operation of the front panel and display

Panasonic[®]

Operating Instructions (Overall)

AC Servo Motor & Driver MINAS A5-series



* This product image is 200W type of A5-series.

- Thank you for purchasing this Panasonic product.
- Before operating this product, please read the instructions carefully, and save this manual for future use.

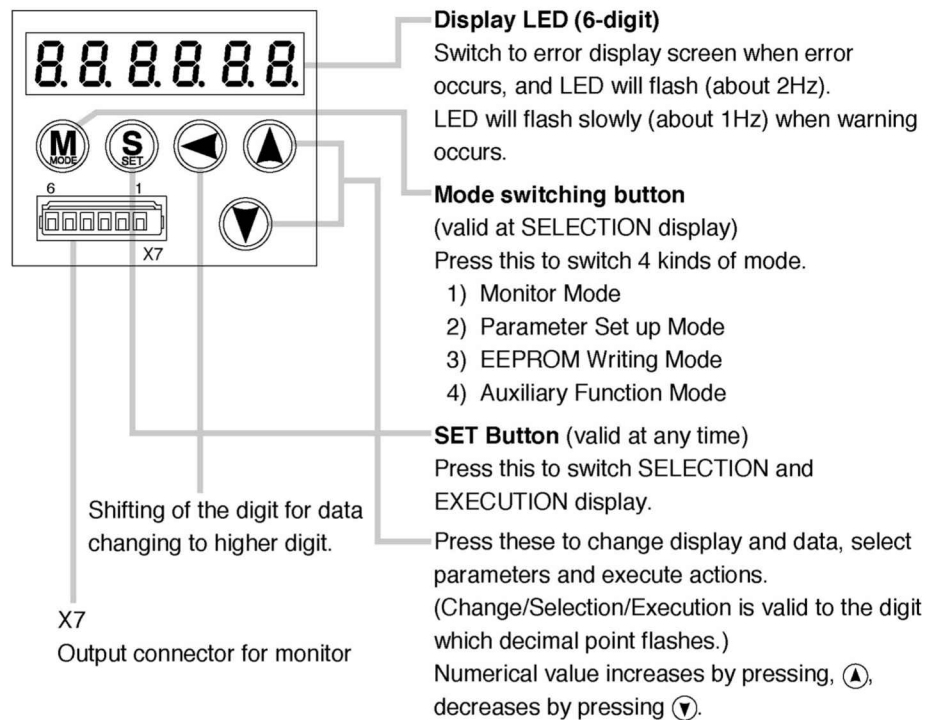
2

Preparation

15. How to Use the Front Panel

Setup

Setup with the Front Panel



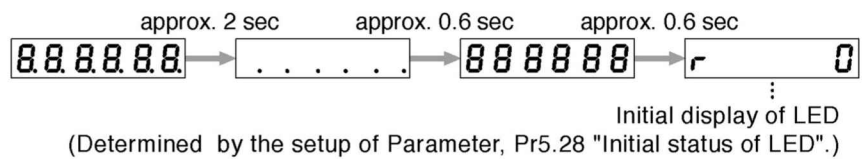
15. How to Use the Front Panel

Setup

Initial Status of the Front Panel Display (7 Segment LED)

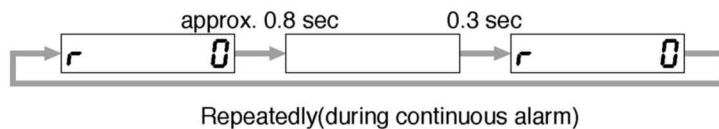
Status

Front panel display shows the following after turning on the power of the driver.



Upon Occurrence of an Alarm

If a driver alarm is generated, the front panel display shows the following repeatedly.



Below shows possible cause of an alarm.

alarm No.	Alarm	Content
A0	Overload protection	Load factor is 85% or more the protection level.
A1	Over-regeneration alarm	Regenerative load factor is 85% or more the protection level.
A2	Battery alarm	Battery voltage is 3.2 V or lower.
A3	Fan alarm	Fan has stopped for 1 sec.
A4	Encoder communication alarm	The number of successive encoder communication errors exceeds the specified value.
A5	Encoder overheat alarm	The encoder detects overheat alarm.
A6	Oscillation detection alarm	Oscillation or vibration is detected.
A7	Lifetime detection alarm	The life expectancy of capacity or fan becomes shorter than the specified time.
A8	External scale error alarm	The external scale detects the alarm.
A9	External scale communication alarm	The number of successive external scale communication errors exceeds the specified value.

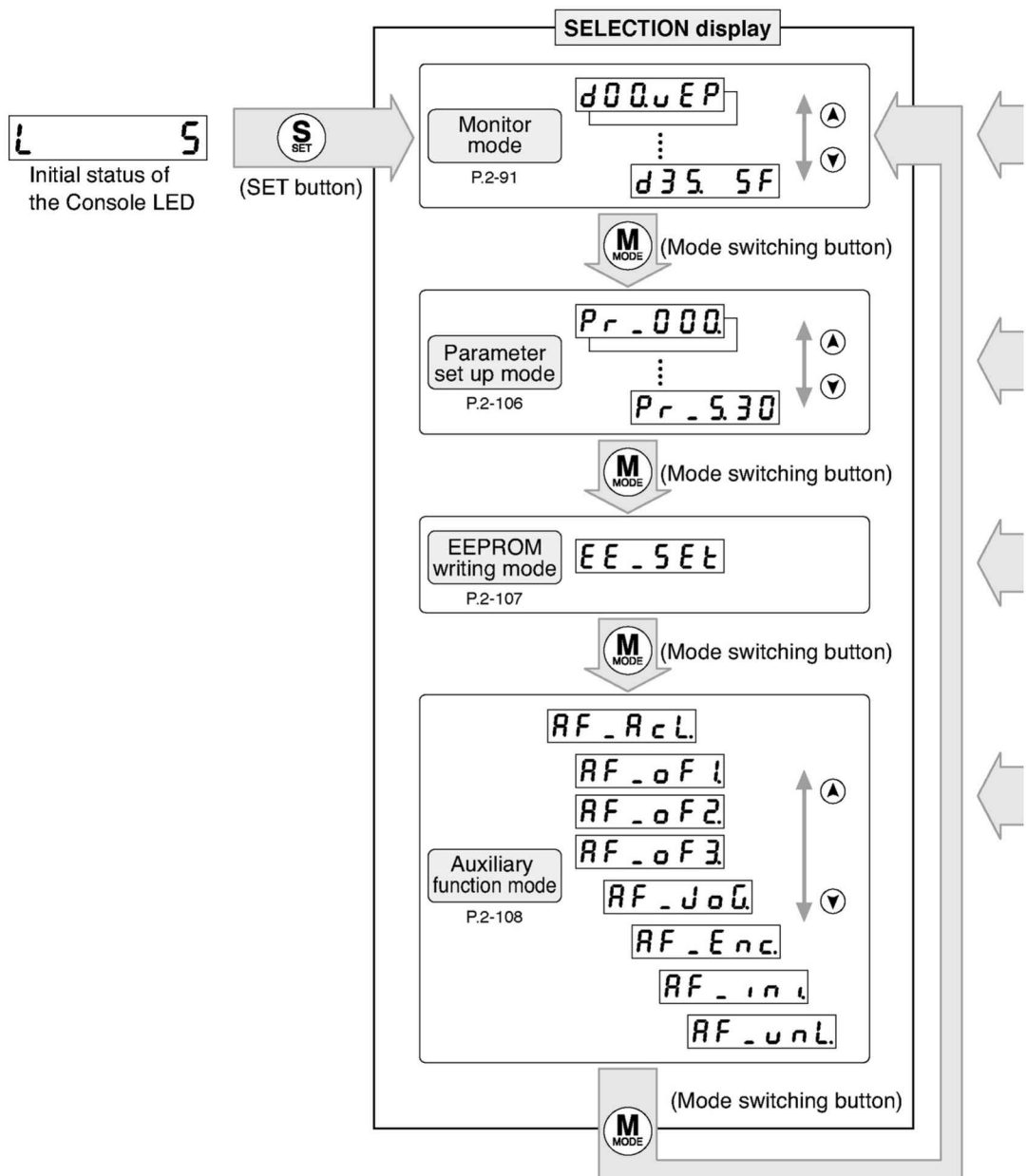
2

Preparation

15. How to Use the Front Panel

Structure of Each Mode

Use each button on the touch panel to select the structure and switch the mode.

**Note**

You can change the flashing decimal point with , then shift the digit for data change “.”

Caution

On power-up, the monitor mode executed is displayed according to the setup of Pr5.28 LED initial status.

15. How to Use the Front Panel

Structure of Each Mode

